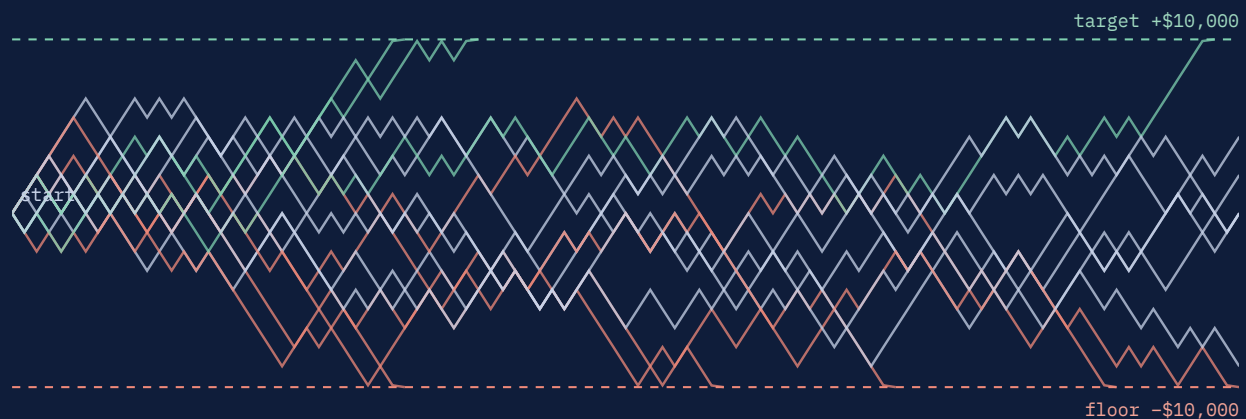


The Prop Firm Option

Why a funded-account challenge has positive expected value for a trader with no predictive skill, exactly how large that value is, how to get it as fast as possible, how far it scales per person, and how to run it inside every firm's rules.



Part 1 · The mathematics, from zero

Part 2 · Execution, step by step

Educational analysis, not financial advice. Prop-firm accounts are simulated and payouts ("rewards") are paid at the firm's discretion under its terms. Every figure in this document assumes the firm pays. In the execution plan about seven attempts in ten lose their full fee; after one month the chance of being ahead is about one in five, and individual results vary enormously. Firms change rules and prices often; check the current terms before acting. Prop firms require traders to be adults (18+). Do not use money you cannot afford to lose.

What is in this document

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- 4 The challenge, step by step
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Key results at a glance

Version 3 re-runs every figure on price paths (Chapter 17), tests trading many positions at once, compares account sizes and every major firm, and adds up the maximum per person. "Zero skill" means no trade direction is better than chance. Every figure assumes the firms pay every payout.

Quantity	Value	Where
Chance a zero-skill trader passes FTMO Phase 1 / both phases (version 3 plan)	46% / 29%	Ch. 3–4, 28
Expected value per attempt, no costs (exact, FTMO 2-Step)	+\$2,235	Ch. 6
Version 3 plan: Nasdaq, stop 0.75 hourly moves, 1 : 5 , risk 1.5%, funded cycles +10%	\$1,520 per attempt, 11.6 days	Ch. 19, 23
... EV per month per FTMO 100K slot (synthetic / real path)	\$4,000 / \$4,500	Ch. 20
... same plan, best and worst CFD firm per slot	\$5,000 (FXIFY) / \$3,600 (GFT)	Ch. 20
Futures firms per 50K account (Topstep best; Apex at list price)	\$1,100 / –\$900	Ch. 20
Several positions at once (Nasdaq + S&P, many pairs)	no gain at the same risk	Ch. 18
100K vs 200K accounts, per dollar of allowance	the same; 50K about 8% worse	Ch. 19
Per person, FTMO only (4 × 100K or 2 × 200K)	\$16,000 per month	Ch. 21
Per person, five best-documented CFD firms (15 slots)	\$68,100 per month	Ch. 21
Per person, seven CFD firms (23 slots)	\$102,400 per month; fees \$32,000	Ch. 21
23 slots: chance ahead after 1 / 3 / 12 months (accounts move together)	36% / 91% / 98%	Ch. 21, 29
23 slots: typical / bad-case (1 in 20) cash low point	–\$76,900 / –\$164,300	Ch. 21, 29

WHAT CHANGED IN VERSION 3

- 1 : 5, not 1 : 3.** Fewer trades, less spread, same time per phase: about 15% more per month.
- Funded cycles of +10%** instead of +\$1,000 first: accounts finish sooner and slots recycle faster.
- One position at a time.** Several positions (correlated or not) never beat one Nasdaq position at the same risk; correlated stacking is a bigger bet in disguise.
- Scale with accounts, not risk.** The per-person maximum comes from running the same plan within the caps of several firms.
- A written direction rule** instead of a coin flip, applied identically everywhere, so no account ever opposes another.
- Figures supersede version 2.** Where Chapters 10–11 differ from Chapters 17–21, the later chapters hold.

PART 1

The mathematics

Everything is built from two facts: a fair game stays fair however you play it, and money paid out has to come from somewhere. No prior knowledge of probability or trading is assumed.

The contract you are buying

A proprietary trading firm ("prop firm") sells evaluations. You pay a fee and receive a **simulated** trading account. Nothing you trade reaches a real market and you can never lose more than the fee. If you meet the firm's targets without breaking its loss limits, you get a "funded" account, still simulated, on which the firm pays you a share of the profits you make. Those payments are called *rewards* or *payouts*.

We use FTMO's 100K 2-Step Challenge as the worked example throughout. Chapter 11 extends the analysis to 37 account types at 16 firms.

Stage	You must reach	You fail if	Other rules
Phase 1 (Challenge)	\$110,000 (+\$10,000)	Equity $\leq$ \$90,000, or you lose \$5,000 in one day	At least 4 trading days; no time limit
Phase 2 (Verification)	\$105,000 (+\$5,000), starting again from \$100,000	Same: floor \$90,000, \$5,000 per day	At least 4 trading days; no time limit
Funded (FTMO Account)	No target	Same floor and daily limit	You keep 80% of withdrawn profit; first payout after 14 days, then every 14 days; the €540 fee is refunded with your first payout

Source: FTMO trading objectives and FAQ, October 2026. The maximum loss is *static*: the floor stays at \$90,000 however high the balance goes.

Words you need

- **Balance** is the account value counting only closed trades. **Equity** also counts the open trade's floating profit or loss. FTMO's limits are checked on equity, in real time.
- **Maximum loss (the floor)**. If equity touches \$90,000 the account is closed. We call the distance from the current balance down to the floor B (it starts at \$10,000).
- **Daily loss limit**. You may not lose more than \$5,000 (5% of the starting balance) within one day, measured from the balance at midnight Prague time, open trades included.
- **Profit target**. The distance from the current balance up to the target is called A.
- **Profit split**. The share of withdrawn profit you keep: 80% at FTMO (90% through its scaling plan).
- **Stop-loss and take-profit**. Orders that close a trade automatically at a loss of S dollars or a profit of T dollars. A trade with both attached is called a **bracket**. Its **reward-to-risk ratio (RR)** is $T : S$.

Why this is an option, and what kind

Your worst case is losing the fee. Your best case has no fixed ceiling. A payoff with a capped downside and an open upside is what finance calls an *option*, and the fee is its price. The question this document answers is whether the option is worth more than its price to someone who cannot

predict the market. Chapters 2 to 6 show that it is, and by how much. Note what this does *not* say. The value does not come from "volatility" in general, and it does not depend on trading in any clever way. Chapter 5 shows exactly where it comes from.

Probability from zero

2.1 Probability and expected value

A **probability** is a number between 0 and 1 saying how often something happens in the long run. A fair coin lands heads with probability $\frac{1}{2}$.

The **expected value** (EV) of a bet is the average result per bet if you repeated it many times. You get it by multiplying each outcome by its probability and adding up.

EXPECTED VALUE

$$EV = p_1 \times outcome_1 + p_2 \times outcome_2 + \dots$$

Example. You win \$10 on heads and lose \$10 on tails: $EV = \frac{1}{2} \times 10 + \frac{1}{2} \times (-10) = 0$. You win \$20 on heads and lose \$10 on tails: $EV = \frac{1}{2} \times 20 - \frac{1}{2} \times 10 = +\5 .

Expected values **add up**. The EV of ten bets is the sum of their ten EVs, whether or not the bets are connected to each other. This one fact will matter a lot in Chapter 13 (hedging).

2.2 Fair games

A bet is **fair** when its EV is zero. A game made of fair bets has a remarkable property: **no way of choosing your bets can change the fact that your expected total is zero**. You can bet big or small, stop when ahead, double after losses. If every individual bet is fair, the expected result of the whole sequence is still zero. Mathematicians call such a process a *martingale*, and the formal version of this fact is the *optional stopping theorem*. Its conditions (bets of bounded size, and a stopping rule that ends the game in finite expected time) are all met by the rules in this document.

2.3 One trade is one bet

Suppose you buy EURUSD and attach a stop 50 pips below and a target 50 pips above. If you have no information about direction, the price is as likely to reach either level first. The bet wins \$T or loses \$S. What is the probability p of winning?

With no skill, the price process has no drift, so the trade is a fair bet before costs: its EV must be zero.

WIN PROBABILITY OF A ZERO-SKILL BRACKET

$$p \times T - (1 - p) \times S = 0 \Rightarrow p = S \div (S + T)$$

Risk : reward	Stop S	Target T	Win probability $S/(S+T)$
1 : 0.1	\$5,000	\$500	90.9%
1 : 1	\$5,000	\$5,000	50.0%
1 : 2	\$5,000	\$10,000	33.3%

Risk : reward	Stop S	Target T	Win probability $S/(S+T)$
1 : 3	\$5,000	\$15,000	25.0%

Higher reward always comes with lower probability, in exactly the proportion that keeps the bet fair. A big target is not "better" and a small one is not "safer". They are the same bet with a different shape.

2.4 Does real data agree?

We tested this on every hour of EURUSD from 18 December 2023 to 2 October 2026. Each test trade picks a random direction and a random entry hour on a weekday, then follows the real price until the stop or target is hit. The results below are before costs.

Stop / target (pips)	Fair value $S/(S+T)$	Measured win rate	95% range
25 / 25	50.0%	50.4%	± 0.7
50 / 50	50.0%	50.3%	± 0.7
55 / 55	50.0%	49.7%	± 0.6
100 / 100	50.0%	50.0%	± 0.7
78 / 39 (1 : 0.5)	66.7%	67.6%	± 0.5
39 / 78 (1 : 2)	33.3%	32.1%	± 0.5
25 / 100 (1 : 4)	20.0%	20.3%	± 0.5

The formula holds closely. Symmetric brackets come out at 50% within noise. There is a small tilt in this sample: far targets (1 : 2) hit about one point less often than fair, and near targets slightly more often, which is mild short-term mean reversion. That is one reason Part 2 uses 1 : 1. The earlier documents claimed an "effective win rate" of 48.5%. The data does not support it. The only systematic drag is the trading cost, which Chapter 7 handles separately.

Reaching a target before a floor

Each challenge phase asks one question: starting from your current balance, will you reach the target (A dollars above) before you hit the floor (B dollars below)? This is the oldest problem in probability, known as the *gambler's ruin*.

3.1 The answer in one line

Let P be the probability of reaching the target first. The game ends in one of two places: $+A$ with probability P , or $-B$ with probability $1 - P$. Every trade is fair, so by Section 2.2 the expected final result must be zero, no matter how you traded:

GAMBLER'S RUIN, EXACT

$$P \times A - (1 - P) \times B = 0 \Rightarrow P = B \div (A + B)$$

For FTMO Phase 1, $A = B = \$10,000$, so $P = 10,000 \div 20,000 = \mathbf{50\%}$. For Phase 2, $A = \$5,000$ and $B = \$10,000$, so $P = 10,000 \div 15,000 = \mathbf{66.7\%}$.

3.2 Checking it by hand

Take a tiny version: steps of \$1, target +2, floor -2. From 0 you need two net wins before two net losses. Call P_x the chance of passing from x . Then $P_2 = 1$, $P_{-2} = 0$, and every middle point is the average of its neighbours: $P_x = \frac{1}{2}P_{x+1} + \frac{1}{2}P_{x-1}$. The only numbers that are "the average of their neighbours" lie on a straight line, so $P_x = (x + 2) \div 4$ and $P_0 = \frac{1}{2}$, exactly as the formula says.

3.3 The value of the phase is a straight line

The same argument works from any balance x between the floor and the target:

PASS PROBABILITY FROM BALANCE x

$$P(x) = (x + B) \div (A + B) \quad (x \text{ measured from the starting balance})$$



Figure 3.1 · Chance of passing Phase 1 from each balance. It is a straight line from 0 at the floor to 1 at the target, whatever the sizing.

At $-\$9,900$ in Phase 1, with only $\$100$ left above the floor, your chance is $100 \div 20,000 = 0.5\%$. You can only risk $\$100$ there, and the formula already accounts for that: it is the room left above the floor that sets the probability.

3.4 What does not change the answer

Look at what the formula does not contain: no bet size, no reward-to-risk ratio, no number of trades, no volatility. **As long as every trade is fair and the account ends exactly on the target or exactly on the floor, the pass probability is $B \div (A + B)$.** Sizing up after losses, sizing down near the floor, 1 : 1 versus 1 : 3: none of it moves the number. Sizing only changes how many trades it takes, and Chapter 7 shows that the number of trades is what you pay for in costs.

This also means the earlier documents' central claim, that the challenge pays off because "variance is rewarded" (Jensen's inequality), is the wrong explanation. With no time limit, more variance does not raise the pass probability at all. Variance only helps when there is a deadline.

3.5 What does change it: overshoot

The formula assumes the account lands exactly on a barrier. If your last winning trade carries you *past* the target, the extra profit is wasted, and the pass probability falls. If your last losing trade carries you *past* the floor (a price gap through your stop), the firm absorbs the extra loss, and the pass probability rises.

GAMBLER'S RUIN WITH OVERTHOOT

$$P = (B + O_{\text{floor}}) \div (A + O_{\text{target}} + B)$$

Example. The earlier documents traded a fixed $\pm\$4,800$ at 1 : 1. Two wins reach only $+\$9,600$, which is short of the $\$10,000$ target, so a third net win is needed and the account effectively has to reach $+\$14,400$. Then $P \approx 10,000 \div (14,400 + 10,000) = 41\%$ instead of 50%. Choosing a fixed size without looking at the target costs nine percentage points.

RULE FROM CHAPTER 3

Size every trade so that a win lands exactly on the target and a loss lands no lower than the floor. Under that rule the pass probability is fixed at $B \div (A + B)$, and the only thing left to optimise is cost.

The challenge, step by step

The cheapest way to land exactly on a barrier is to use as few trades as possible. Within FTMO's \$5,000 daily loss limit that means risking up to \$5,000 per trade and aiming the take-profit straight at the target. Mathematicians call this **bold play**. Here is the complete tree.

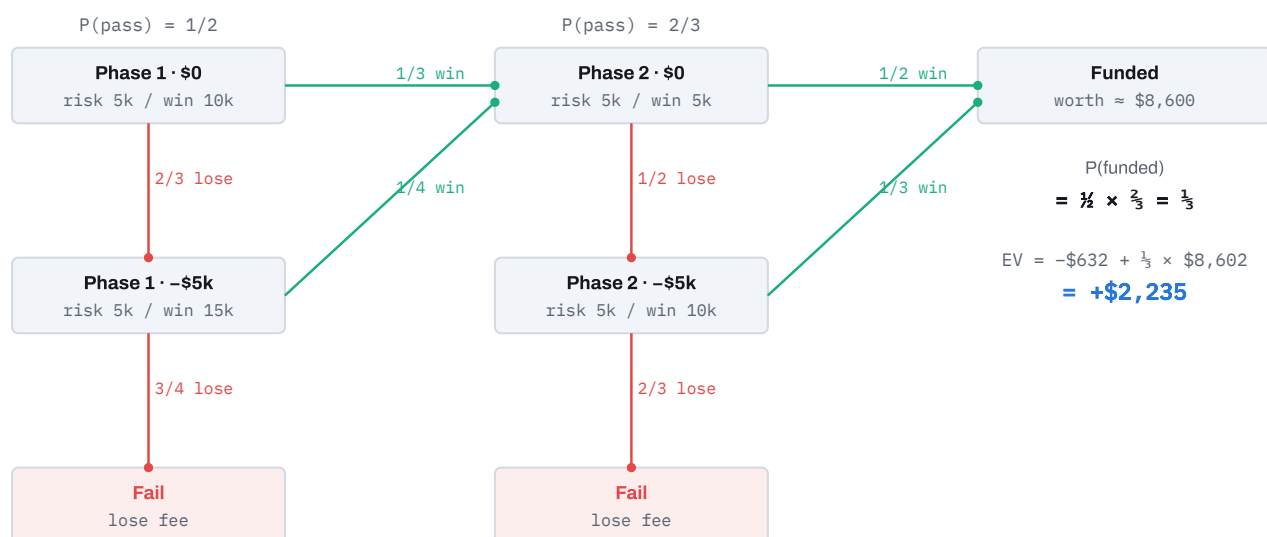


Figure 4.1 · Bold play through both phases. Each trade risks the daily limit and aims exactly at the target.

4.1 Phase 1

Where you are	Trade	Win chance	If it wins	If it loses
Day 1, at \$0	Risk \$5,000, target \$10,000 (1 : 2)	1/3	Pass	Balance -\$5,000
Day 2, at -\$5,000	Risk \$5,000, target \$15,000 (1 : 3)	1/4	Pass	Floor: fail

$$P(\text{pass Phase 1}) = 1/3 + (2/3) \times (1/4) = 1/3 + 1/6 = 1/2$$

Check it with plain 1 : 1 trades of \$5,000 instead. Two wins in a row pass (1/4). Two losses in a row fail (1/4). One of each brings you back to \$0 (1/2), where you start again. So $P = 1/4 + 1/2P$, which gives $P = 1/2$. It is the same answer, as Chapter 3 promised. Bold play just gets there in at most two trades.

4.2 Phase 2

Where you are	Trade	Win chance	If it wins	If it loses
Day 1, at \$0	Risk \$5,000, target \$5,000 (1 : 1)	1/2	Pass	Balance -\$5,000
Day 2, at -\$5,000	Risk \$5,000, target \$10,000 (1 : 2)	1/3	Pass	Floor: fail

$$P(\text{pass Phase 2}) = 1/2 + (1/2) \times (1/3) = 2/3 \quad P(\text{funded}) = 1/2 \times 2/3 = 1/3$$

4.3 Details the tree hides

- **Minimum trading days.** If you pass on day 1 or 2, you still need 4 trading days. Place a tiny trade (for example 0.01 lots with a 5-pip stop) on the remaining days. It changes nothing measurable.
- **The daily limit includes open losses.** Keep each stop a little under \$5,000 (the real-data test used \$4,850) so spread and slippage cannot breach the limit.
- **Phases are independent.** Phase 2 starts again from \$100,000, so its 2/3 does not depend on how Phase 1 went.
- **This is the theoretical optimum, not the Part 2 plan.** Risking the full daily limit on one trade aimed at the target is what FTMO's conduct rules call overexposure or a one-sided bet (Chapter 17). Part 2 keeps risk at 1.5% per trade and gets its speed from the instrument and the stop size instead (Chapters 8–10).

What a funded account is worth

Once funded, there is no target. You trade, withdraw profits every 14 days, keep 80%, and the account continues until equity hits the \$90,000 floor. How much do you expect to collect in total?

5.1 An accounting identity

Follow the money. Every dollar withdrawn from the account came either from trading profit or from the balance going down. Over the whole life of the account:

FUNDED-ACCOUNT IDENTITY (ALWAYS TRUE)

$$Total\ withdrawn = Total\ trading\ profit + (Starting\ balance - Final\ balance)$$

The account ends when the balance reaches the floor, so $Starting - Final = \$10,000$. With zero skill, total trading profit is a sum of fair bets, whose expected value is zero (Section 2.2; the stopping conditions hold because the account is reset to \$100,000 at every payout, which guarantees it ends in finite expected time). Taking expectations:

EXPECTED PAYOUT OF A FUNDED ACCOUNT

$$E[total\ withdrawn] = 0 + \$10,000 = \$10,000 \Rightarrow \text{you receive } 80\% \times \$10,000 = \$8,000 \text{ (minus costs)}$$

In words: **the firm's expected payout equals the loss allowance it gives you.** You keep 80% of every dollar you withdraw, and the firm pays every dollar of loss down to the floor. In a fair game, the dollars won balance the dollars lost, and the losses stop at exactly \$10,000. This is the true source of the option's value. It does not depend on how you trade.

5.2 Simulated check: trading style does not matter

Funded-account trading style	Avg paid, no costs	Avg paid, 1.4% costs	Chance of any payout
Bold: risk \$5k, aim +\$500 first, then +\$5k cycles	\$8,779 ± 137	\$8,340	95%
1% risk, 1 : 1, every day	\$8,579 ± 136	\$7,080	79%
5% risk, 1 : 2, every day	\$8,171 ± 241	\$7,541	27%

20,000 simulated funded accounts per row. Each figure is \$8,000 plus the refunded fee times the chance of at least one payout (the refund only arrives with a first payout). Costs reduce the payout most for the style that trades most.

5.3 Why the earlier "expected maximum = \$10,000" claim is false

The earlier documents argued that a fair walk with a floor \$10,000 below reaches an expected maximum of \$10,000. The chance that it ever climbs m dollars before the floor is $10,000 \div (10,000 + m)$, which falls so slowly that the expected maximum is *infinite*. The correct reason the payout

averages \$10,000 is the identity in 5.1, which depends on withdrawing profits. It also shows that their simulated averages of \$10,271 (which would mean \$12,839 withdrawn) and $13.6\% \times \$22,000$ (which would mean \$8,985 to you per funded account) are impossible with zero skill.

5.4 The refund and the first payout

The €540 fee comes back with your first payout, so it pays to make the first payout very likely. Aim first for a small profit h . The chance of reaching $+h$ before the floor is $10,000 \div (10,000 + h)$. With $h = \$500$ that is 95.2%. Once you reach $+\$500$, stop trading until the payout date, collect \$400 plus the refund, then continue normally. The refund adds about $0.95 \times \$632 \approx \textbf{\$600}$ to the funded value, for a total of about **\$8,600**.

Putting it together

VALUE OF ONE ATTEMPT

$$EV = -\text{Fee} + P(\text{pass 1}) \times P(\text{pass 2}) \times \text{Funded value}$$

$$= -\$632 + \frac{1}{2} \times \frac{2}{3} \times \$8,602 = -\$632 + \$2,867 = +\$2,235$$

6.1 Every possible outcome of one attempt

With bold play in the funded account, each payout cycle after the first pays \$4,000 with probability 2/3 and ends the account with probability 1/3 (the same tree as Phase 2). That makes the full distribution exact:

Outcome	Probability	Net result
Fail Phase 1	50.0%	-\$632
Pass Phase 1, fail Phase 2	16.7%	-\$632
Funded, but bust before the first payout	1.6%	-\$632
First payout (\$400 + refund), then bust	10.6%	+\$400
... plus one \$4,000 cycle	7.1%	+\$4,400
... plus two cycles	4.7%	+\$8,400
... plus three cycles	3.1%	+\$12,400
... plus four or more	6.3%	+\$16,400 and up
Expected value	100%	+\$2,235



Figure 6.1 · One FTMO 100K attempt under bold play, no costs. The expected value is positive, but the most likely single outcome is losing the fee.

The EV is large, but the typical outcome is a loss: 68% of attempts lose the whole fee. The value sits in a long right tail. That shape is why Chapter 12 (bankroll) matters as much as the EV itself.

6.2 If it is this good, why does the firm survive?

On a perfect zero-skill trader, FTMO expects to pay out about \$2,870 for every \$632 fee. Firms survive for three reasons.

- **Most customers are far worse than random.** Industry estimates put real Phase 1 pass rates at 10–15%, against 46–50% for a low-cost coin flip. The difference is mostly trading costs from many small trades with tight stops (Chapter 7), plus abandoning live accounts.
- **Rules remove convexity where it is cheap to remove:** trailing drawdowns, consistency rules, payout caps (Chapter 11).
- **Payouts are discretionary**, and the conduct rules target exactly the fastest version of the strategy (Chapter 16).

Trading costs

Every trade pays the **spread** (the gap between buying and selling price), a **commission**, and sometimes **slippage** (being filled worse than your stop price). Costs are the only thing in this model that turns fair bets into losing ones.

7.1 Measuring cost per trade

For EURUSD on FTMO, a realistic all-in cost is about 0.75 pip per round trip: roughly 0.15 pip of raw spread, 0.5 pip of commission (about \$5 per standard lot) and 0.1 pip of average slippage. What matters is the cost *relative to the stop*. We call this ratio κ (kappa):

COST RATIO

$$\kappa = \text{cost in pips} \div \text{stop distance in pips} \quad \text{e.g. } 0.75 \div 55 \approx 1.4\%$$

A trade risking \$1,000 with a 55-pip stop costs about \$14. The same \$1,000 risk with a 10-pip stop needs 5.5 times the position size, so it costs about \$75 ($\kappa = 7.5\%$).

7.2 Costs as a drift

A cost of c dollars per trade turns the fair bet into one with $EV = -c$. The pass formula becomes, approximately:

PASS PROBABILITY WITH COSTS

$$P \approx (B - \text{expected total cost}) \div (A + B)$$

7.3 Why the number of trades is what you pay for

With fair trades of size L (risk L , target L), the expected number of trades to finish a phase is $A \times B \div L^2$. Each costs about $\kappa \times L$, so:

EXPECTED TOTAL COST OF A PHASE

$$\text{Cost} \approx \kappa \times L \times (A \times B \div L^2) = \kappa \times A \times B \div L$$

Phase 1 with $\kappa = 1.4\%$	Risk per trade L	Expected trades	Expected cost	P(pass)
Bold play (Chapter 4)	\$5,000	≈ 1.7	$\approx \$180$	48.9%
Slow 1% plan (version 1)	\$1,000	≈ 100	$\approx \$1,400$	$\approx 44\text{--}45\%$
Scalper, 10-pip stops ($\kappa = 7.5\%$)	\$500	≈ 400	$\approx \$15,000$	$\approx 0\%$

The scalper's expected costs exceed the whole loss allowance. This one line explains most of the gap between the textbook 50% and the industry's 10–15%.

7.4 How to keep costs low

- **Wide stops are cheap but slow.** A stop of about one day's range keeps κ near 1–1.5% on EURUSD, but each trade then lasts days. Chapter 11 shows how to trade cost against time.
- **Liquid instruments and hours.** EURUSD during the London and New York sessions has the tightest spreads.
- **Avoid scheduled news.** Rate decisions, US payrolls and CPI cause slippage on stops.
- **Swap.** Holding overnight earns or pays a small interest charge. With ~1–2 day trades it is minor, but check your platform's swap table.

With $\kappa \approx 1.4\%$ applied to every trade, the simulated EV of the bold plan falls from +\$2,272 to **+\$2,081** (100,000 simulated attempts).

Time: how long an attempt takes

Expected value per attempt is only half the story. If one plan earns \$1,400 per attempt and needs 86 weeks, and another earns \$1,400 and needs 3 weeks, the second is worth almost thirty times more per month. This chapter derives how long each phase takes and which choices make it faster.

8.1 How many trades a phase needs

Take fair trades that risk L and aim for $k \times L$ (reward-to-risk $1 : k$). Each trade adds a random amount with mean zero and variance $k \times L^2$. A second fair-game identity (Wald's) says the expected squared final result equals the expected number of trades times the variance per trade. The final result is $+A$ with probability $B \div (A + B)$ and $-B$ otherwise, so its expected square is $A \times B$.

EXPECTED NUMBER OF TRADES IN A PHASE

$$N = A \times B \div (k \times L^2)$$

FTMO Phase 1 ($A = B = \$10,000$) with 1% risk and $1 : 1$ needs about 100 trades. With 1.5% risk and $1 : 3$ it needs about $10,000 \times 10,000 \div (3 \times 1,500^2) \approx 15$ trades.

8.2 How long one trade takes

A bracket closes when price moves a distance s against you or $k \times s$ in your favour. If the price moves with volatility σ per hour (its typical one-hour move), the same identity applied to price gives:

EXPECTED DURATION OF ONE BRACKET

$$T_{trade} \approx k \times s^2 \div \sigma^2 \quad \text{hours}$$

A stop of one hourly move ($s = \sigma$) at $1 : 1$ takes about one active hour. A stop of three hourly moves takes about nine.

8.3 Time and cost of a whole phase

Multiply the two. The reward-to-risk ratio k cancels:

PHASE TIME AND PHASE COST

$$Time \approx A \times B \times s^2 \div (L^2 \times \sigma^2) \quad Cost \approx A \times B \times c \div (k \times L \times s)$$

Here c is the round-trip cost as a fraction of price. These two lines contain every lever:

Lever	Effect on time	Effect on cost	Limit
Risk per trade L	Falls with L^2 : 1.5% is 2.25× faster than 1%	Falls with L	Firm guidance: FTMO suggests 1–1.5%
Stop distance s	Rises with s^2	Falls with s	The trade-off you optimise
Reward-to-risk k	No effect	Falls with k	Overshoot near the target; real-data drift at far targets
Instrument	Sets the price of speed: cost c against movement σ		Chapter 9
Leverage	None, unless it forces a wider stop than you want (Section 9.3)		Rarely binds

8.4 The speed–cost trade-off

A tighter stop finishes the phase sooner but costs more per dollar risked. If the expected value per attempt falls roughly in a straight line with cost, $EV \approx EV_0 - \beta/s$, and time grows like s^2 , then EV per unit time is $(EV_0 - \beta/s) \div s^2$. Setting its derivative to zero gives $s = 1.5 \beta \div EV_0$, where EV is one third of EV_0 .

THE SURPRISING RULE

When time is what matters, the best stop is tight enough that trading costs eat about **two thirds** of the no-cost expected value. Fixed waiting times (minimum days, reviews, the 14-day payout calendar) push the optimum back towards wider stops, because once those waits dominate, more speed buys little. Chapter 10 finds the real optimum by simulation.

8.5 Fixed waits you cannot trade away

- **Minimum trading days:** 4 per phase at FTMO, 3 at FundingPips and The5ers, 5 at FundedNext.
- **Reviews:** about 1 day between phases and 1–3 days for identity checks and the contract before funding.
- **The payout calendar:** FTMO's first payout comes 14 days after your first funded trade, then every 14 days. FundingPips offers weekly payouts at a 60% split. FundedNext's first cycle is 21 days. Crypto firms such as Breakout and HyroTrader pay on demand.

These waits set a floor. Even the fastest pass cannot produce an FTMO payout much sooner than about 25 days after purchase: 4 + 4 minimum trading days, about 4 days of reviews, then 14 days to the first payout date.

Instruments, leverage and platforms

9.1 The one number that ranks instruments

From Section 8.3, for a fixed time budget, the cost you pay is proportional to $c \div \sigma$: the trading cost compared with how far the instrument moves. Over a full trading week this becomes:

INSTRUMENT EFFICIENCY

Efficiency = typical weekly move $\div$ round-trip trading cost (higher = faster for the same cost)

Weekly movement already includes trading hours. An instrument that trades 24/7 gets credit for its weekend movement automatically, so "crypto trades all week" is already counted.

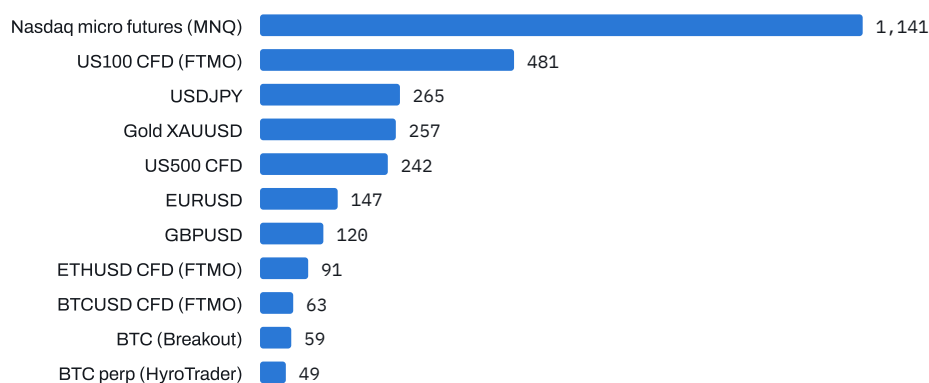


Figure 9.1 · Weekly move divided by round-trip cost. Higher means an instrument can be traded faster for the same cost.

Instrument (platform)	Weekly move	Round-trip cost	Efficiency	Max leverage	Hours/week
Nasdaq 100 micro futures (futures firms)	3.08%	0.0027%	1,141	–	~110
US100 CFD (FTMO, zero commission)	3.08%	0.0064%	481	1:50	~110
USDJPY (FTMO)	1.35%	0.0051%	265	1:100	~118
Gold XAUUSD (FTMO)	2.78%	0.0108%	257	1:30	~110
US500 CFD (FTMO)	2.32%	0.0096%	242	1:50	~110
EURUSD (FTMO)	0.91%	0.0062%	147	1:100	~118
GBPUSD (FTMO)	0.91%	0.0076%	120	1:100	~118
ETHUSD CFD (FTMO, 0.0325%/side commission)	9.52%	0.105%	91	1:3.3	168
BTCUSD CFD (FTMO)	5.89%	0.093%	63	1:3.3	168
BTC (Breakout/Kraken, 0.04%/side)	5.89%	0.10%	59	1:5	168
BTC perp (HyroTrader/Bybit, taker 0.055%/side)	5.89%	0.12%	49	up to 1:100	168

Weekly move: standard deviation of weekly returns, hourly data 2023–26. Costs: published FTMO spreads (US100 1.66 pts, gold \$0.52, EURUSD raw spread plus \$3/lot commission), exchange fees for crypto firms, plus an allowance for slippage. Futures: about \$1.70 per micro contract round trip. Check your own platform: Chapter 18.

9.2 Crypto: trading 24/7 does not help

Bitcoin moves about six times as much as EURUSD in a week, partly because it trades on weekends. But on prop platforms it costs fifteen to twenty times as much per trade, because crypto carries percentage commissions (0.0325% per side at FTMO, 0.04% at Breakout, 0.055% taker at Bybit). Its efficiency ends up between a half and a tenth of the alternatives. Chapter 10 confirms it: the best crypto set-up found earns about a fifth of the Nasdaq plan per month.

9.3 What leverage actually does

Leverage does not make trades resolve faster. Time depends on how far price must travel (the stop distance) compared with how fast it moves. Leverage only limits position size, and so sets the *tightest* stop you can use for a given risk:

TIGHTEST STOP ALLOWED BY LEVERAGE λ

$$s_{min} = \text{risk per trade} \div (\lambda \times \text{account size}) \quad 1.5\% \text{ risk at } 1:50 \rightarrow 0.03\%; \text{ at } 1:3.3 \rightarrow 0.45\%$$

For indices, metals and currencies the limit is far below any sensible stop. For crypto at FTMO (1:3.3), 0.45% is just above one hourly Bitcoin move, so leverage would block the fastest crypto stops. Costs make those stops unattractive anyway.

9.4 Overnight financing

Positions held through the daily rollover pay (or occasionally earn) financing on their full size. With tight stops the position is large, so even one night can cost several percent of the risk on index and gold CFDs. Fast plans close most trades within hours, which keeps this small. The simulations charge it on every rollover crossed.

9.5 Futures

Futures are the cheapest instrument of all, but futures prop firms use trailing drawdowns, small accounts (\$2,000 allowance on 50K), payout ladders and consistency rules (Chapter 11). Their value per account is small, though attempts are quick and some firms allow many accounts at once.

Maximising expected value per month

VERSION 3 NOTE

This chapter is version 2's analysis. Chapters 17–21 redo it on price paths and find a better plan (1 : 5, funded cycles of +10%) and higher values per month. Where the figures differ, Chapters 17–21 hold.

10.1 Method

For each instrument we measured, on real hourly data, how long random-direction brackets of each size actually take (including waiting for trading hours and weekends) and how often they cross the rollover. Those measured durations then drive a full simulation of each firm's contract with calendar time:

- Daily loss limits, minimum days and reviews.
- Payout calendars and refunds.
- A daily guard that stops trading before the daily limit can be breached.

Every set-up was simulated 1,200–4,000 times. The trades are assumed to have **no edge**: the small win-rate deviations seen in the data were treated as noise. The result is *expected value per month per account slot* = expected value per attempt ÷ average length of an attempt.

10.2 Stop size

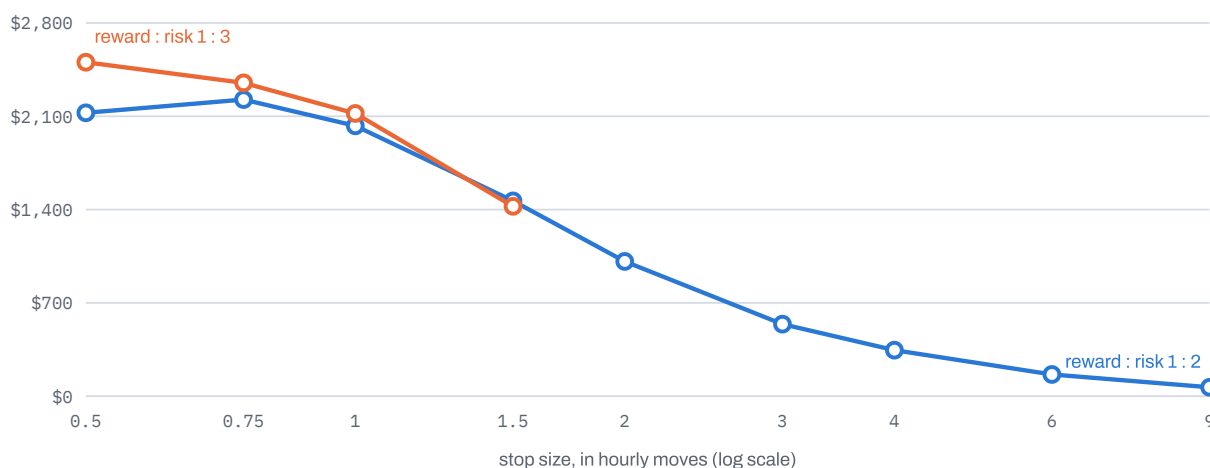


Figure 10.1 · EV per month per slot against stop size. FTMO 2-Step, US100, 1.5% risk, zero edge. The best stops are short.

The best stop is short: about **0.5–1 hourly move**, which is 40–85 Nasdaq points. At 0.75 hourly moves, costs take roughly 40% of the no-cost value, which matches the trade-off of Section 8.4 once the fixed payout waits are included. Very wide stops (the earlier plan used about six hourly moves on EURUSD) are thirty times slower.

10.3 Reward-to-risk, risk per trade and payout policy

FTMO 2-Step, US100	EV per attempt	Days per attempt	EV per month
Stop 1 hourly move, 1 : 1, risk 1.5%	+\$880	22	\$1,236
Stop 1 hourly move, 1 : 2, risk 1.5%	+\$1,469	22	\$2,029
Stop 1 hourly move, 1 : 3, risk 1.5%	+\$1,402	20	\$2,122
Stop 0.75 hourly moves, 1 : 3, risk 1.0%	+\$1,234	25	\$1,503
Stop 0.75 hourly moves, 1 : 3, risk 1.5%	+\$1,355	18	\$2,351
Same, funded target +\$3,000 per cycle instead of +\$10,000	+\$1,526	25	\$1,892

- **Reward-to-risk 1 : 2 or 1 : 3** beats 1 : 1 by a wide margin. As Section 8.3 predicts, it costs no time and cuts the number of trades.
- **1.5% risk** beats 1%, also as predicted (time falls with L^2). Both are inside FTMO's 1–1.5% guidance; this plan does not go above it.
- **Funded payout policy.** Aim for a large profit per 14-day cycle (+\$10,000) rather than a small one. Fewer cycles mean the account resolves sooner. The first cycle still aims for +\$1,000 to secure the fee refund.

10.4 Instrument

FTMO 2-Step, best set-up per instrument	EV per month per slot
US100 (Nasdaq CFD)	\$2,350–2,500
Gold (XAUUSD)	\$2,071
USDJPY	\$1,814
US500	\$1,467
EURUSD	\$1,179
GBPUSD	\$781
ETHUSD CFD	\$299
BTCUSD CFD	\$83

10.5 Firm

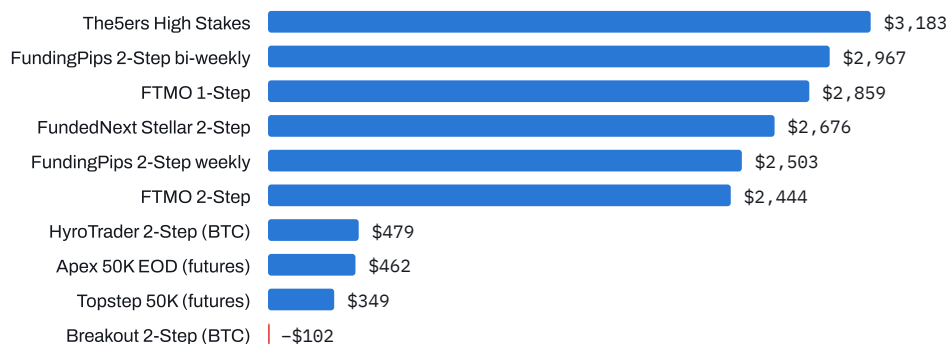


Figure 10.2 · EV per month per account slot, best set-up found for each programme (zero edge, payouts honoured).

Firm and programme (100K, US100 plan)	EV per attempt	Days per attempt	EV per month	Median days to first payout
The5ers High Stakes	+\$1,869	18	\$3,183	23
FundingPips 2-Step, bi-weekly 80%	+\$1,743	18	\$2,967	23
FTMO 1-Step	+\$1,755	19	\$2,859	23
FundedNext Stellar 2-Step	+\$1,866	21	\$2,676	34
FundingPips 2-Step, weekly 60%	+\$1,171	14	\$2,503	16
FTMO 2-Step	+\$1,429	18	\$2,444	27
Apex 50K EOD (futures, MNQ, \$500 risk)	+\$148	10	\$462	13
HyroTrader 2-Step (BTC, 3% risk)	+\$571	36	\$479	28
Topstep 50K (futures, MNQ, \$500 risk)	+\$76	7	\$349	12
Breakout 2-Step (BTC)	-\$250	75	-\$102	78

±\$170–210 on EV per attempt for the CFD firms (4,000 simulations each). The same Nasdaq costs were assumed at every CFD firm; their real spreads may differ. Futures values are per 50K account.

The CFD firms with an 8% Phase 1 target (The5ers, FundingPips) come out ahead of FTMO 2-Step's 10%. The FTMO 1-Step's 90% split and resetting floor also beat it. The weekly 60% payout option pays sooner but less. The crypto firms and Breakout's 6% drawdown with a 10% Phase 2 target are poor contracts.

10.6 Check on real prices

We re-ran the winning FTMO set-up on the actual 2024–26 Nasdaq price path, with every bracket resolved on real bars:

Real-path test, FTMO 2-Step (1,500 attempts each)	EV per attempt	Days	EV per month
US100, stop 0.75 hourly moves, 1 : 3, 1.5% risk	+\$2,076 ± 372	19	\$3,281
Same, trading costs × 1.5	+\$1,596 ± 322	18	\$2,688
US100, stop 1 hourly move, 1 : 2	+\$1,675 ± 355	22	\$2,289

Real-path test, FTMO 2-Step (1,500 attempts each)	EV per attempt	Days	EV per month
EURUSD, stop 1 hourly move, 1 : 3	+\$1,214 ± 282	18	\$2,004

Real prices came out above the zero-edge model. Part of that is favourable noise: in this sample, far targets were hit slightly more often than fair odds. Plan with the zero-edge numbers, about **\$2,400 per month per FTMO 2-Step slot**.

10.7 What one account slot looks like over a year

One FTMO 100K slot, fast plan, rebuying after each attempt	1 month	3 months	6 months	12 months
Chance of being ahead	5%	36%	69%	88%
Average result	-\$1,757	+\$1,088	+\$7,781	+\$23,110
Median result	-\$1,896	-\$1,560	+\$5,344	+\$20,584
Bad case (10th percentile)	-\$3,160	-\$5,056	-\$6,616	-\$1,272
Good case (90th percentile)	-\$632	+\$13,008	+\$23,872	+\$49,648

2,000 simulated years. Fees are counted when paid; payouts are counted at the end of each attempt (slightly pessimistic timing). About 1.6 attempts per month per slot, so roughly \$1,000 a month in fees.

Other firms and other rule types

VERSION 3 NOTE

This chapter is version 1's firm comparison. Chapters 17–21 redo it on price paths and find a better plan for every major firm (Chapter 20). Where the figures differ, Chapters 17–21 hold.

Every challenge can be valued with the same two tools: a pass probability for each phase, and the identity for the funded account. What changes between firms is the shape of the floor and the payout rules.

11.1 Static floor

FTMO 2-Step and most forex/CFD 2-Steps work as in Chapters 3–5: $P = B \div (A + B)$ per phase, funded value $\approx \text{split} \times B$.

11.2 End-of-day (EOD) trailing drawdown

The floor rises whenever a day closes at a new high, and stays there. If you can pass in a single day, the floor never moves and you get the static answer. Firms add **consistency rules** ("your best day may not exceed X% of total profit") to force several profitable days, each of which lifts the floor.

11.3 Intraday trailing drawdown

The floor follows your highest *live equity*, including unrealised profit. With prices that move continuously, the best possible chance of reaching A before a drawdown of D from the peak is:

INTRADAY TRAILING, ANY STRATEGY

$P = e^{-A/D}$ (for driftless prices, the peak reached before a drawdown of D is exponentially distributed with mean D)

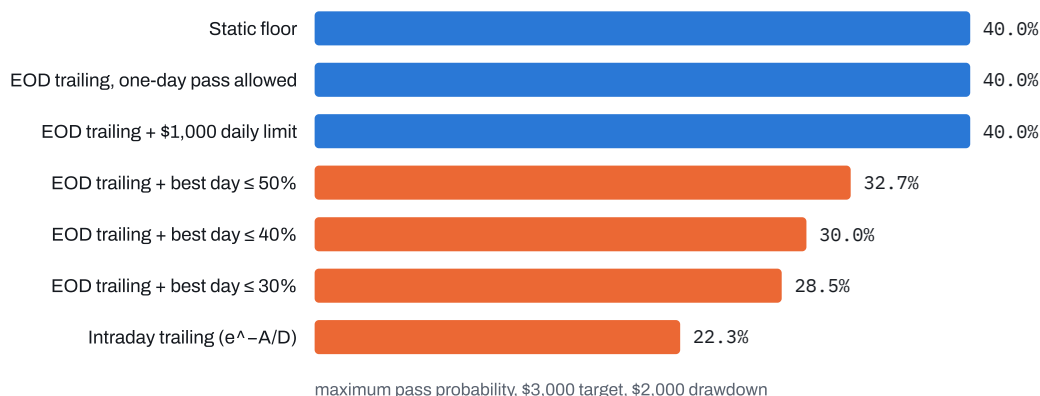


Figure 11.1 · How each rule type changes the best achievable pass probability (dynamic programming; intraday is the continuous-price bound).

11.4 Funded accounts with locks and caps

Futures firms typically start the funded floor at the same trailing distance, then "lock" it at the starting balance plus about \$100 once you are far enough ahead. They also cap payouts (for example Apex closes the account after six payouts). The identity still holds, so the funded value is $\text{split} \times (\text{expected start} - \text{final balance})$. Locks and caps reduce it, typically to around \$1,500 on a \$2,000 allowance, and to about \$700 when the trail is intraday.

11.5 The ranking

We valued 37 account types at 16 firms the same way: zero skill, best sizing for each rule set, 1% trading costs, payouts honoured. The full table is in Appendix B.

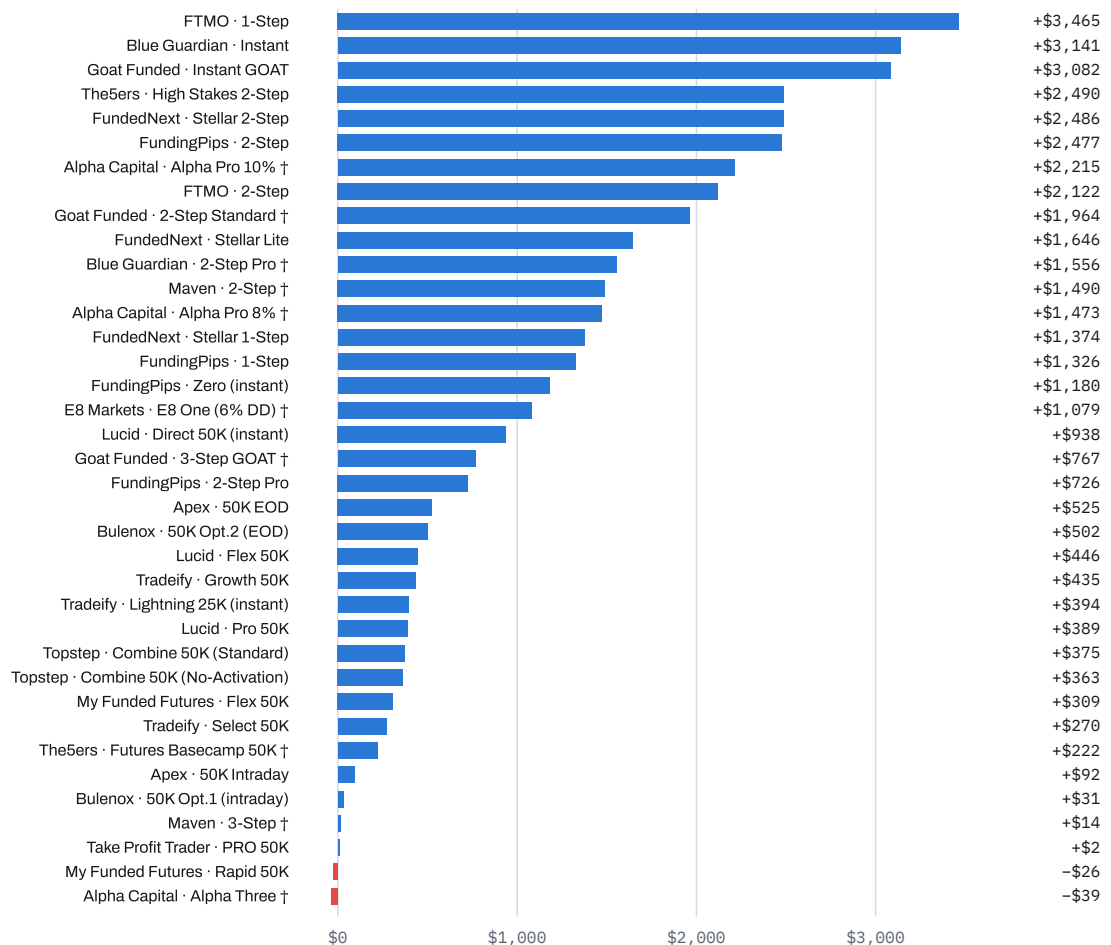


Figure 11.2 · Expected value per attempt, zero skill, 1% costs, payouts honoured. † rule set partly inferred.

- **Static 10% 2-Steps** (FundedNext, FundingPips, The5ers, FTMO): +\$2,000 to +\$2,500 per attempt.
- **FTMO 1-Step** ranks first (+\$3,465): its trailing floor resets with each payout and the split is 90%.
- **Futures evaluations** are cheap and return the most per dollar of fee, but each is worth only \$300–\$550.
- **Intraday trailing** accounts and **3-step** programmes are worth close to nothing, or less.
- **Instant-funding** values are upper estimates: their payout caps and consistency rules were not fully modelled.

Variance, bankroll and Kelly

A positive EV only shows up after many attempts. Because most attempts lose the fee and a few pay a lot, you need enough money to survive the losing runs.

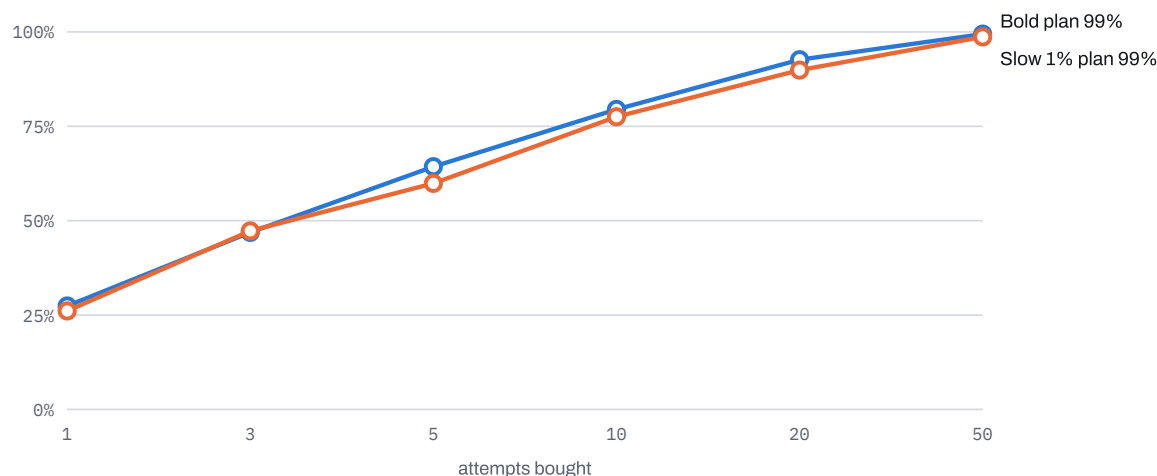


Figure 12.1 · Chance of being ahead after buying n FTMO 100K attempts (real EURUSD data, bootstrap).

Attempts bought	1	3	5	10	20	50
Chance of being ahead, bold plan	27%	47%	64%	79%	93%	99%
Chance of being ahead, slow 1% plan	26%	47%	60%	78%	90%	99%

Bootstrap from 6,000 (bold) and 3,000 (slow 1% plan) simulated attempts on real EURUSD data. The fast plan of Part 2 is shown month by month in Section 10.7.

12.1 Risk of ruin

If you start with a fixed amount and buy attempts until you either run out of fees or complete 100 attempts, the chance of running out (bold plan, real data) is 53% with 3 fees (\$1,896), 35% with 5 fees (\$3,160), 11% with 10 fees (\$6,320) and 1.2% with 20 fees (\$12,640). The slow 1% plan is riskier per attempt, because more of its attempts lose: 22% ruin with 10 fees and 5% with 20.

12.2 The Kelly criterion, done correctly

The Kelly criterion picks the bet size that makes your wealth grow fastest over time. For a bet with mean gain μ and variance σ^2 per unit staked, the growth-optimal stake is roughly $\mu \div \sigma^2$ of your wealth. For a fixed-size bet such as one challenge, this inverts: the bankroll at which buying one attempt is growth-optimal is about $\sigma^2 \div \mu$.

BANKROLL FOR ONE ATTEMPT AT FULL KELLY

$W \approx \sigma^2 \div \mu$ exact log-growth on the real-data outcomes: $\approx \$4,100$ (bold) and $\approx \$6,000$ (slow 1% plan)

The earlier documents computed $\mu \div \sigma^2 = 1.4 \times 10^{-5}$, treated it as a fraction of wealth, and concluded you need \$40 million. The units are wrong: $\mu \div \sigma^2$ is "per dollar", not a percentage. Half-Kelly (about \$8,000–12,000) is a sensible minimum for someone who wants smaller swings.

Hedging

The idea is to take opposite positions in two accounts, at the same firm, at two firms, or against a personal brokerage account, so that one account's loss is the other's gain.

13.1 Hedging cannot create value

Expected values add (Section 2.1). The expected total of several accounts is the sum of their individual EVs, however their results are linked. If each account is worth +\$2,235, a hedged pair is worth exactly +\$4,470. If each were worth −\$100, hedging would lock in −\$200. A hedge changes only the *spread* of outcomes.

13.2 The values are straight lines, so a hedge can be exact

With a static floor, the value of each stage is a straight line in its own balance (Section 3.3):

Stage (no costs)	Value at start	Value per \$1 of balance (slope)
Funded account	\$8,602	0.80
Phase 2	\$5,735	$8,602 \div 15,000 = 0.573$
Phase 1	\$2,867	$5,735 \div 20,000 = 0.287$

Holding an opposite position equal to the slope makes the total value constant. For example, two Phase 1 accounts with opposite positions (equal target and floor distances) always produce exactly one pass, and a personal account holding $0.287\times$ the opposite exposure locks in \$2,867 − \$632 per attempt. Mathematically, hedging turns the lottery of Chapter 6 into a near-certain cash flow.

13.3 Why it does not work in practice

- **It is prohibited.** FTMO's forbidden-practices page bans opposite positions across accounts, explicitly including "accounts held with various operators/providers", with forfeiture of rewards and termination of all agreements. Most firms have equivalent clauses.
- **It cannot protect against a refused payout.** Against your own brokerage account, the paths where the prop account wins are exactly the paths where your real money lost. If that payout is refused, the loss is real, up to \$8,000 per funded account, where the unhedged loss would only have been the fee.

Hedging is therefore covered here only to explain the mathematics. It is not part of the execution plan.

Testing it on real market data

14.1 Method

Data: hourly EURUSD bars, 18 December 2023 to 2 October 2026 (730 trading days). Each simulated attempt starts on a random date and opens each trade at a random weekday hour between 07:00 and 16:00 UTC, in a random direction. Brackets resolve on real highs and lows. When one hourly bar touches both stop and target, a coin flip decides (this cannot bias the result). Costs are 0.75 pip per round trip. FTMO's daily and maximum loss limits, minimum days, 14-day payout cycles, 80% split and refund are all applied. Funded accounts run until they hit the floor. When the data runs out it wraps to the start.

14.2 Results

Strategy	EV per attempt	P(pass 1)	P(funded)	P(profit)	Avg length
Bold play (Chapter 4), 6,000 attempts	+\$1,764 ± 161	46.4%	29.5%	28.0%	≈ 6 weeks
Slow 1% plan (version 1): 1% risk, 1 : 1, 3,000 attempts	+\$1,398 ± 193	45.0%	27.2%	26.1%	≈ 86 weeks
Same with 1 : 2 targets	+\$1,368 ± 209	38.2%	23.2%	22.2%	≈ 44 weeks
Earlier PDFs' protocol (±\$4,800, 1 : 1, quit after 2 losses, daily flatten)	-\$173 ± 79	14.6%	8.2%	3.5%	–

± values are 95% confidence ranges. "Avg length" counts market time from purchase to the end of the attempt (failure, or the funded account hitting its floor).

The real-data figures sit a little below the cost-only model (+\$2,081 for bold play). The gap comes from the safety buffer kept above the floor, hourly resolution, and the mild mean reversion seen in Section 2.4. The sign and size are confirmed. The fast plan of Part 2 was tested the same way in Section 10.6.

14.3 Why the earlier protocol loses

Earlier protocol variant	EV per attempt	P(funded)
As written: quit Phase 1 after 2 full losses, flatten before New York close	-\$70 ± 90	8.8%
Quit rule kept, no daily flatten	+\$404 ± 165	14.8%
No quit rule, daily flatten kept	+\$884 ± 152	22.1%
Neither	+\$672 ± 178	19.9%

The quit rule is the main damage. A live account is always worth $(x + B) \div (A + B)$ of the next stage's value (Section 3.3), so abandoning it throws that value away. After a win-win-loss-loss sequence, for instance, you are back at \$0 with a full 50% chance, and the rule tells you to quit. The fixed ±\$4,800 size adds the overshoot loss of Section 3.5.

Errors in the earlier "Convex Edge" documents

For readers coming from the earlier "FTMO Convex-EV" / "Convex Edge" PDFs, this is what changes and why.

Earlier claim	Correct statement	Status
The edge comes from variance via Jensen's inequality; EV rises with σ^2 .	With no deadline, the pass probability is $B/(A+B)$ for any σ . The edge is the loss allowance (Chapter 5).	wrong mechanism
A fair walk's expected maximum before the floor equals the floor distance.	It is infinite. The funded value follows from the withdrawal identity.	false
Funded payout \$10,271 at 80% split; EV +\$3,223 / +\$2,502.	Implies \$12,839 withdrawn against a \$10,000 allowance, which is impossible. The EV formula also refunded the fee to every funded account, including those that never got paid.	impossible
13.6% $\times$ \$22,000 naive EV.	\$8,985 per funded account, above the \$8,000 ceiling.	impossible
1:1 is the only RR that makes a symmetric walk.	Every fair bracket is a martingale. RR matters only through overshoot.	false
Fixed $\pm$ \$4,800 trades at 1:1.	Overshoots the target and cuts Phase 1 from 50% to 41%.	costly
A constant $\sim$ \$30 cost per trade regardless of size creates a "trap" near the floor.	Costs scale with lot size. With proportional costs there is no trap, and dynamic sizing does not change the pass probability.	model bug
Effective win rate of 48.5% (spread asymmetry plus mean reversion).	Measured 49.7–50.4% for symmetric brackets. The spread asymmetry is the spread cost, so it was counted twice.	not in data
Quit Phase 1 after two consecutive full losses.	Throws away live value. On real data, removing it moves the protocol from $-\$70$ to $+\$884$.	main loss
Kelly implies a \$40M bankroll.	Units error. About \$4,100 for bold play, \$6,000 for the Part 2 plan.	10,000 $\times$ off
RR table shows 1:1.5 as best.	The table's own maximum was 1:0.5; the differences are simulation noise.	noise
Use 1:2 on the funded account for more EV.	Funded value is $\text{split} \times B$ for any RR.	irrelevant
Abandoned accounts free up allocation.	They remain open and count toward the cap.	false
"Defensibility": vary lots, rotate sessions, add decoy trades, stagger purchases.	These disguise a pattern the terms prohibit. If discovered, they justify forfeiture. Part 2 follows the rules instead.	removed
\$4,600–5,000 per month from four accounts.	Derived from wrong per-attempt maths with a losing protocol. A corrected fast plan inside the risk guidance (Chapter 13) reaches a similar order, about \$2,400 per FTMO slot per month in the zero-edge model, if payouts are honoured.	wrong derivation
Zero-skill play is positive EV if payouts are honoured.	Correct, and by more per attempt than the earlier documents said.	correct

What the mathematics cannot tell you

Every number so far assumes the firm pays what it owes. That assumption is the weakest link, and no model can price it.

- **Payouts are discretionary.** Funded accounts are simulated, and "rewards" are paid under terms that let the firm refuse them for practices it considers non-genuine.
- **Conduct rules.** FTMO lists "overleveraging, overexposure, one-sided bets, account rolling" and hedging across accounts as forbidden, and recommends risking 1–1.5% per trade. The bold plan of Chapter 4 matches the prohibited pattern.
- **Firm failure.** About 80–100 prop firms closed in 2024–25. True Forex Funds closed owing about \$1.2 million; The Funded Trader admitted to more than \$2 million in denied payouts; MyForexFunds was shut by a CFTC action in 2023.
- **Rule changes.** Fees, splits and limits change often. Each change moves the numbers in this document.
- **Behaviour.** The maths assumes you follow the plan exactly through long losing runs. Skipping trades, sizing up to "win it back" or quitting live accounts all lower the result.
- **Model limits.** The real-data test covers 2.8 years of one currency pair. A different period could show more or less mean reversion. Costs differ between platforms.

BREAK-EVEN PAYOUT REFUSAL RATE (FTMO 2-STEP, 1% COSTS)

$EV \geq 0$ as long as fewer than about 77% of owed payouts are refused

That looks comfortable, but refusals are not independent coin flips. A refusal usually comes with the account closed and the person barred, which ends all future attempts. The realistic protection is to trade in a way that does not breach the terms at all. That is the purpose of Part 2.

Version 3: every trade resolved on a price path

Version 2 modelled each account with a calendar and the firm's rules, but drew each trade's outcome and duration from distributions measured once on real data. Version 3 removes that shortcut. Every trade is opened at a real hour, its stop and target are checked bar by bar against an hourly price path, and the firm's rules (daily loss, floors, minimum days, payout dates, caps, refunds) are applied around it.

17.1 Two kinds of price path

Path	What it is	Used for
Synthetic	A random walk with exactly zero edge, the Nasdaq's real hourly volatility, 12 years long, markets closed at weekends. Highs and lows inside each hour are drawn from the exact distribution of a continuous path (a "Brownian bridge"), so stops are monitored as real stop orders are.	All headline figures. A zero-skill trader can do no better or worse than this, on average.
Real	Hourly Nasdaq-100 bars, 2024–26, with measured spreads and overnight financing.	A check. One two-year path carries its own trends, so real-path figures scatter around the synthetic ones.

17.2 Checks that the engine is fair

Check (synthetic path, no costs)	Theory	Engine
Win rate of a 1 : 1 bracket	50.0%	49.98%
Win rate of a 1 : 2 bracket	33.3%	33.4%
Average result of a 1 : 3 bracket, in units of risk	0	+0.013 ± 0.020
FTMO Phase 1 pass rate, one position at a time	49.6%	49.2%
FTMO Phase 1 pass rate, four positions at a time	49.6%	49.9%

17.3 Four traps found on the way

Each of these made a simulation look better than reality. All four are fixed in every figure of version 3, and each one is also a trap for anyone writing their own test.

1. **Skipping the entry bar.** Checking the stop and target only from the next hour onwards ignores what the price did in the first hour. It handed a zero-skill 1 : 3 trade a fake +0.14 R per trade.
2. **Checking the price too rarely.** Sampling the path 12 times an hour misses stop touches between samples. The nearer level (the stop) is missed more often, which faked +0.11 R per trade.
3. **Overshooting the target with several positions.** When one position hits the phase target, the others are still open; their profit beyond the target is wasted. Without a cap on the combined

target, the pass rate with four positions fell from 49.6% to 43%, while the average balance at a pass rose to +\$13,665 (Chapter 18).

- A take-profit equal to the distance left.** Costs are charged on the whole position, so a "win" of the last \$40 can still lose money and the account never lands. The target must be the distance left *plus* the trade's cost (already a rule in version 2).

17.4 What changed compared with version 2

FTMO 2-Step 100K, zero edge	EV per month per slot	Average attempt
Version 2 account model (1 : 3, funded +\$1,000 then +\$10,000)	\$2,444	–
Version 3 path engine, same settings	\$3,272	15.3 days
... with +10% funded cycles	\$3,634	11.6 days
... and 1 : 5 instead of 1 : 3 (the version 3 plan)	\$3,998	11.6 days
Version 3 plan on the real 2024–26 path	\$4,527	16.2 days

Three things moved the number. Trades on a real path take about 3.6 hours including waiting, not the 7 hours version 2 assumed, so attempts are shorter. Funded accounts are run in +10% cycles instead of +\$1,000 first (Chapter 19). And a 1 : 5 target beats 1 : 3, because it needs fewer trades and so pays less spread (Chapter 19).

Many positions at once

The idea: instead of one trade at a time on one instrument, open several at once (Nasdaq and S&P 500, or many pairs), so the account reaches its target or floor sooner. This chapter works out what that can and cannot do, then tests every variant on price paths.

18.1 Three facts

1. **The pass chance does not change.** A fair game stays fair however many bets run at once, so a phase still passes with probability $B \div (A + B)$. That holds only if profit beyond the target is not wasted (18.3).
2. **Correlated positions in the same direction are one bigger bet.** Two positions risking L_1 and L_2 on instruments with correlation ρ carry the risk of a single position of $\sqrt{(L_1^2 + L_2^2 + 2\rho L_1 L_2)}$. Nasdaq and S&P 500 move together with $\rho \approx 0.9$, so 1.5% on each is about 2.9% on one idea. That is twice FTMO's 1–1.5% guidance on what is really one trade. It also costs more than a single 2.9% Nasdaq position, because the S&P CFD costs about twice as much per unit of movement (Chapter 9).
3. **Independent positions add speed, but the calendar does not move.** Two independent positions double the rate at which the balance moves, so trading time halves. But an attempt also contains minimum trading days, account reviews and payout dates, which no amount of trading shortens.

18.2 Results

Set-up (FTMO 2-Step 100K)	Open risk	EV per month	vs one position	Days per attempt	Pass both phases
Nasdaq alone, 1.5%	one position	\$3,180	+0%	15.1	29%
Nasdaq + S&P 500, same direction, 0.75% each	total 1.5%	\$2,301	-28%	15.2	26%
Nasdaq + S&P 500, same direction, 1.5% each	$\approx 2.9\%$ on one idea	\$3,570	+12%	13.3	28%
Nasdaq alone, 3% (reference, beyond guidance)	one position	\$3,993	+26%	12.6	30%
Nasdaq + gold, 1.5% each, at most 1.5% open	one at a time	\$3,067	-4%	15.4	29%
Nasdaq + gold, 1.5% each	up to 3% open	\$2,681	-16%	13.9	27%
Nasdaq + gold + yen + euro, 1.5% each	up to 4.5% open	\$2,727	-14%	15.5	27%
Two independent Nasdaq-like markets, 1.5% each (imaginary)	up to 3% open	\$3,220	+1%	14.7	29%

Set-up (FTMO 2-Step 100K)	Open risk	EV per month	vs one position	Days per attempt	Pass both phases
Four independent Nasdaq-like markets, 1.5% each (imaginary)	up to 6% open	\$3,315	+4%	13.6	29%
Nasdaq alone, no costs at all	one position	\$4,129	+30%	16.9	32%
Four independent markets, no costs at all	up to 6% open	\$4,614	+45%	16.6	34%

Synthetic zero-edge paths, 6,000 attempts per row, combined target capped. These runs used the version 2 settings (1 : 3, funded +\$1,000 first), so compare rows with each other rather than with later chapters. Noise is about $\pm$ \$330 per month per row.

Read the table in three groups.

- **Same total risk, split across correlated instruments** (Nasdaq + S&P at 0.75% each): 28% worse than one Nasdaq position at 1.5%. The risk is the same, so there is no speed gain, and the spread is paid twice.
- **Twice the risk, via a correlated second position** (1.5% each): 12% better than one 1.5% position, but 11% worse than simply trading one Nasdaq position at 3%. Stacking is the expensive way to take a bigger bet, and either way the bet exceeds the guidance.
- **Independent instruments** (gold, yen, euro next to the Nasdaq): worse, because every other instrument costs two to three times more per unit of movement. Even an imaginary second Nasdaq with independent prices and identical costs adds only 1–4%, within the noise, because the calendar dominates.

18.3 The overshoot trap

With several positions open, the target can be crossed by more than one of them. Profit beyond the target is pass probability thrown away: by optional stopping, $P(\text{pass}) = (B + \text{overshoot at the floor}) \div (A + B + \text{overshoot at the target} + \text{overshoot at the floor})$. Four uncapped positions (no costs) passed Phase 1 only 43% of the time, against the theoretical 49.6%, because the average balance at a pass was +\$13,665 rather than +\$10,000. Capping the *combined* target of all open positions at the distance left, and closing everything once equity reaches the target, restored 49.9%.

VERDICT

One position at a time, on the cheapest instrument per unit of movement, at the top of the firm's guidance. More positions help only if a second instrument is as cheap as the Nasdaq and independent of it, and none is. The way to grow income is more accounts, not more positions per account (Chapter 21).

Account size, account type and the settings that matter

19.1 50K, 100K or 200K

Account	Fee	Fee ÷ size	EV per month	per \$100K of allowance
FTMO 1-Step 100K	\$584	0.58%	\$3,248	\$3,248
FTMO 1-Step 200K	\$1,168	0.58%	\$6,748	\$3,374
FTMO 2-Step 50K	\$404	0.81%	\$1,668	\$3,336
FTMO 2-Step 100K	\$632	0.63%	\$3,634	\$3,634
FTMO 2-Step 200K	\$1,264	0.63%	\$7,545	\$3,773

1 : 3, funded cycles of +10% (1-Step: +5% first), 12,000 synthetic attempts each; noise about ±4% per row.

The value of an account is proportional to its loss allowance, and so is everything else, including the risk per trade. What differs is the fee per dollar of allowance: FTMO charges about 0.63% of the account size for 100K and 200K, but 0.81% for 50K. So, per dollar of the \$400,000 cap, two 200K accounts and four 100K accounts are worth the same, and 50K accounts about 8% less. Four 100K accounts give smoother results and can be started at different times; two 200K accounts mean fewer logins and orders.

19.2 1-Step or 2-Step

FTMO's 1-Step (one 10% target, 90% split, but a 3% daily limit, an end-of-day trailing floor and a 50% best-day rule) gives \$3,700 per month per slot, against \$4,000 for the 2-Step. The best-day rule forces profits to be spread over several days, and the trailing floor shrinks the allowance after good days. The 2-Step is better for this plan.

19.3 The settings, one at a time

FTMO 2-Step 100K, synthetic paths, zero edge. Each row changes one setting from the plan (stop 0.75 hourly moves, 1 : 5, risk 1.5%, funded cycles of +10%). Rows marked with an asterisk used 1 : 3 because they were run before 1 : 5 was adopted; compare them with the 1 : 3 baseline beside them.

Setting (all else as in the plan)	EV per month	Days per attempt	Pass both phases
Baseline at 1 : 3 (stop 0.75, risk 1.5%, +10% cycles)*	\$3,634	11.6	29%
Risk 1% per trade*	\$2,790	14.5	27%
Risk 2% per trade* (beyond FTMO guidance)	\$4,303	10.9	30%
Risk 3% per trade* (beyond FTMO guidance)	\$5,080	9.9	30%
Stop 0.5 hourly moves*	\$3,509	10.0	27%

Setting (all else as in the plan)	EV per month	Days per attempt	Pass both phases
Stop 1 hourly moves*	\$3,104	12.9	29%
Stop 1.5 hourly moves*	\$2,987	19.0	32%
Stop 3 hourly moves*	\$1,180	45.7	31%
Funded: first cycle +\$1,000, then +\$10,000*	\$3,272	15.3	30%
Funded: first cycle +\$3,000, then +\$10,000*	\$3,095	13.5	29%
Funded: first cycle +\$5,000, then +\$10,000*	\$3,563	13.0	29%
Funded: first cycle +\$10,000, then +\$20,000*	\$3,610	11.1	29%
Reward : risk 1 : 1*	\$1,758	11.4	24%
Reward : risk 1 : 2*	\$2,778	11.1	27%
Reward : risk 1 : 4 (real path: \$4,298)	\$3,725	11.5	30%
Reward : risk 1 : 5 (real path: \$4,334)	\$4,269	12.0	30%
Reward : risk 1 : 6 (real path: \$4,101)	\$4,245	11.9	30%
Reward : risk 1 : 8 (real path: \$4,012)	\$4,389	12.0	30%
Reward : risk 1 : 10 (real path: \$4,029)	\$4,195	12.0	31%
Reward : risk 1 : 15 (real path: \$4,020)	\$4,022	12.1	30%
Reward : risk 1 : 3 on the real path	\$3,701	15.5	31%

12,000 synthetic attempts per row (seed 11); noise about $\pm \$300$ per month. The final figure for the 1 : 5 plan, from 24,000 attempts with other seeds, is \$4,000 (Chapter 20): the same within noise.

19.4 Why 1 : 5 beats 1 : 3

Trades per phase fall in proportion to the reward-to-risk ratio k ($A \times B \div k L^2$), and so does the spread paid. Time per trade rises in proportion to k , so the time per phase barely changes. The gain stops around 1 : 5 to 1 : 8, because near a target most trades are capped at the distance left anyway. On the real 2024–26 path the best was 1 : 4 to 1 : 5. Version 3 uses 1 : 5.

The price of 1 : 5 is psychological. Only about one trade in six wins. A run of ten losses in a row has a probability of about $(5/6)^{10} \approx 16\%$ at any given point and will happen many times a year. Nothing is wrong when it does.

19.5 Funded accounts: run +10% cycles

With zero edge, the total paid out by a funded account averages the split times its loss allowance, whatever the payout policy (Chapter 5). What the policy changes is time. A small first target (+\$1,000) reaches the first payout safely, but then the account sits idle for days waiting for the date. Aiming for +10% in every cycle finishes accounts sooner and recycles the slot into a new attempt. At FTMO (1 : 3): \$3,634 per month with +10% cycles against \$3,272 with +\$1,000 first.

Every firm, compared on the same paths

Each programme below was run with the plan of Part 2 (Nasdaq, stop 0.75 hourly moves, 1 : 5, risk 1.5% of the account; for futures, the best of a grid of settings), 24,000 attempts on synthetic paths and 8,000 on the real path. Rules and prices were checked in October 2026 (Appendix D).

20.1 CFD firms (100K accounts)

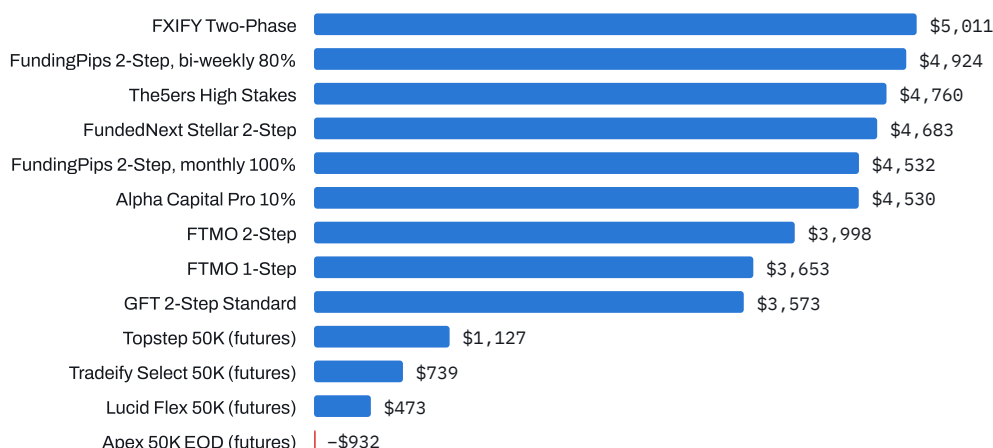


Figure 20.1 · EV per month per slot, version 3 plan, synthetic zero-edge paths (CFD slots are 100K, futures 50K).

Programme (100K)	Fee	Pass	Paid per funded acct	Funded days	EV per attempt	Days per attempt	EV / month	Real path	1st payout (day)
FXIFY Two-Phase 10/5%, refund with 1st payout	\$480	30%	\$7,223	16	\$1,659	10.1	\$5,011	\$5,042	22
FundingPips 2-Step, bi-weekly 80% 8/5%, refund with 4th payout	\$529	33%	\$7,187	17	\$1,871	11.6	\$4,924	\$4,933	24
The5ers High Stakes 10/5%, 3 profitable days	\$491	30%	\$7,326	16	\$1,705	10.9	\$4,760	\$4,886	24
FundedNext Stellar 2-Step 8/5%, 5 days, +15% of challenge profit, first payout day 21	\$550	34%	\$8,106	19	\$2,178	14.2	\$4,683	\$4,922	35
FundingPips 2-Step, monthly 100% 8/5%, 100% split, monthly	\$529	34%	\$8,759	30	\$2,417	16.2	\$4,532	\$4,540	40
Alpha Capital Pro 10% 10/5%, 3 days, no refund assumed	\$497	30%	\$7,149	17	\$1,644	11.0	\$4,530	\$4,606	24
FTMO 2-Step 10/5%, 4 days per phase, 14-day payouts	\$632	29%	\$7,302	16	\$1,520	11.6	\$3,998	\$4,527	26

Programme (100K)	Fee	Pass	Paid per funded acct	Funded days	EV per attempt	Days per attempt	EV / month	Real path	1st payout (day)
FTMO 1-Step 10%, 3% daily, trailing floor, best day ≤ 50%	\$584	33%	\$6,829	16	\$1,664	13.9	\$3,653	\$3,538	24
GFT 2-Step Standard 10/5%, funded profit ≤ \$3,000/day, first 2 payouts ≤ 6%	\$524	30%	\$6,772	25	\$1,500	12.8	\$3,573	\$3,374	22

The CFD firms differ by less than one would think: between \$3,600 and \$5,000 per month per slot, about ±\$170 of simulation noise each. The real path agrees within its noise. What separates them is mostly how long each step takes (minimum days, first payout date) and side rules: GFT caps funded profit at \$3,000 a day, FundingPips refunds the fee only with the fourth payout, FundedNext pays 15% of challenge profits with the first payout and waits 21 days for it.

20.2 Futures firms (50K accounts, micro Nasdaq futures)

Programme (50K)	Fee	Pass	Paid per funded acct	Funded days	EV per attempt	Days per attempt	EV / month	Real path
Topstep 50K risk \$950, 1 : 3	\$49/month + \$149	25%	\$817	6	\$117	3.2	\$1,127 ±235	\$1,727
Tradeify Select 50K risk \$950, 1 : 5	\$165	22%	\$1,138	6	\$84	3.5	\$739 ±329	\$970
Lucid Flex 50K risk \$950, 1 : 3	\$146	25%	\$783	5	\$48	3.1	\$473 ±274	\$994
Apex 50K EOD risk \$500, 1 : 5	\$490	37%	\$504	19	-\$304	9.9	-\$932 ±81	-\$145

Futures programmes are much weaker for this plan. Their loss allowance is small (\$2,000 on 50K, against \$10,000 on a 100K CFD account), it trails the balance, it locks just above the starting balance, and payouts need five qualifying days and are capped at about half the profit. Each funded account pays out only about \$500–1,100 on average. Topstep is the strongest; Tradeify and Lucid are smaller and less certain. Apex at its \$490 list price loses money: at \$147 (a 70% discount) it would be \$39 per attempt, about \$120 per month; at \$98 (a 80% discount) it would be \$88 per attempt, about \$270 per month. Only the price decides whether Apex is worth running, and even then it is small.

20.3 Crypto programmes

Unchanged from Chapter 9: 24/7 trading does not compensate for crypto's cost per unit of movement, which is 5–10 times the Nasdaq's. Leverage only sets the tightest stop possible; at 1.5% risk it does not bind on any instrument except FTMO's crypto CFDs.

The maximum per person

Every firm limits how much it allocates to one person. Within those limits, the plan can run in parallel at several firms. This chapter adds it up.

21.1 The caps

Firm	Allocation rule (October 2026)	Slots used	Source quality
FTMO	\$400,000 per trader or strategy across all accounts, 1-Step and 2-Step combined. Treated as including evaluations.	4 × 100K	A
FundingPips	\$400,000 across Evaluation, Master and Prime accounts (evaluations count).	4 × 100K	A
FundedNext	\$300,000 in funded accounts; evaluations not limited.	3 × 100K	A
The5ers	High Stakes: one 50K or 100K account (plus smaller ones).	1 × 100K	A
Alpha Capital	\$400,000 per household; \$300,000 per strategy.	3 × 100K	B
FXIFY	Reported as \$805,000 in total, one active account per size. Confirm with support.	4 × 100K (assumed)	C
GFT	\$400,000 in funded accounts.	4 × 100K	B
Topstep	5 Express Funded accounts at a time.	5 × 50K	A
Apex / Tradeify / Lucid	20 / 5 / not stated. Not used: negative or marginal value (Chapter 20).	—	B

Source quality: A = the firm's own help pages or several consistent sources; B = consistent secondary sources; C = one secondary source, to be confirmed. Details in Appendix D.

A "slot" is one account that is kept running: when an attempt ends (failed evaluation or lost funded account), the next one is bought. Where a firm counts evaluations in its cap (FTMO is treated this way to be safe; FundingPips says so explicitly), the number of slots is the cap divided by the account size. Futures firms cap funded accounts, not evaluations. Firms not modelled (E8 Markets reports a \$4.85 million household cap; dozens of smaller firms exist) could add more in theory, but their rules could not be checked to the same standard.

21.2 Expected value per month, per person

Portfolio (each slot runs the version 3 plan)	Slots	EV per month	Real path	Fees per month
FTMO only: 4 × 100K	4	\$16,000	\$18,100	\$6,600
Five best-documented CFD firms: 15 slots	15	\$68,100	\$71,300	\$21,200
Seven CFD firms: 23 slots	23	\$102,400	\$105,000	\$32,000
Seven CFD firms + Topstep: 23 + 5	28	\$108,100	\$113,600	\$36,200

"Fees per month" is what the slots spend on evaluations (and activations) at their usual pace; payouts arrive from the second month. The real-path figure checks the synthetic one on 2024–26 prices. Adding Tradeify and Lucid (five 50K accounts each) would add roughly \$6,100 per month in theory, with large uncertainty (Chapter 20).

21.3 Why all accounts move together

No two of your accounts may hold opposite positions at the same time; at most firms that would be hedging, and FTMO also forbids opposite positions across firms. The simplest way to obey that is for every account to trade the same direction at the same time, which makes their results move together. A bad week is bad on every account at once. The table therefore shows two cases: "together" (every slot does as well or as badly as the others, the realistic case) and "independent" (as if each slot had its own luck, which this plan cannot achieve).

Portfolio	Ahead at 1 month	3 months	12 months	Median after 12 months	Bad case (1 in 20)	Typical cash low	Bad-case cash low
FTMO only together	19%	75%	96%	\$167,500	\$17,600	-\$15,200	-\$55,600
FTMO only independent (not achievable)	30%	88%	100%	\$173,400	\$91,300	-\$3,400	-\$15,800
Five best-documented CFD firms together	34%	88%	98%	\$717,300	\$157,900	-\$51,800	-\$120,200
Five best-documented CFD firms independent (not achievable)	22%	100%	100%	\$735,500	\$581,800	-\$11,800	-\$33,000
Seven CFD firms together	36%	91%	98%	\$1,101,900	\$243,700	-\$76,900	-\$164,300
Seven CFD firms independent (not achievable)	26%	100%	100%	\$1,125,800	\$930,000	-\$11,700	-\$39,300
Seven CFD firms + Topstep together	35%	91%	98%	\$1,143,800	\$187,200	-\$95,700	-\$215,300
Seven CFD firms + Topstep independent (not achievable)	23%	100%	100%	\$1,180,200	\$980,700	-\$14,200	-\$42,600

2,000 simulated 12-month histories per firm. "Together": all slots at the same rank of luck. "Cash low": the lowest point of cumulative cash (fees paid minus payouts received).

READ THIS BEFORE THE TOTALS

These are expected values under three assumptions: zero skill, every firm pays every payout, and no firm changes its rules. Payouts are discretionary. Running the same short, systematic trades on many accounts at seven firms is exactly what risk teams review. One refusal usually closes the account and bars the person. Several prop firms have failed or stopped paying since 2023. The totals are an upper limit of what the contracts are worth, not a forecast of income.

21.4 If evaluations were not capped

FundedNext states that its \$300,000 cap applies to funded accounts only and that evaluations may be bought without limit. In principle, extra evaluations can keep every funded slot full; each FundedNext funded slot is then worth about \$10,200 per month instead of about \$4,700. In practice this needs evaluations whose results do not move together, and the same-direction rule makes them move together. If several pass at once, the extras cannot be funded. The headline therefore uses only capped slots.

21.5 Growing to the maximum

Start with one firm and add the next only from payouts already received. The order puts the best-documented caps and longest payout records first.

Stage	Accounts	Slots	EV per month	Fees per month
1	FTMO 2-Step × 4	4	\$16,000	\$6,600
2 (after first payouts)	+ FundingPips × 4, The5ers × 1	9	\$40,400	\$13,600
3	+ FundedNext × 3, Alpha Capital × 3	15	\$68,100	\$21,200
4 (caps confirmed)	+ FXIFY × 4, GFT × 4	23	\$102,400	\$32,000

Direction rules, and what one price history can tell you

The plan assumes no skill, so the direction of each trade does not affect expected value as long as the rule is not systematically wrong. Version 2 used a random choice. Version 3 recommends a written mechanical rule instead (Section 25.3). A risk team can recognise a mechanical rule as a trading plan; FTMO lists "trading without a plan" under the gambling it monitors. Four rules were run on the real 2024–26 Nasdaq path:

Rule (FTMO 2-Step, 1 : 3, real 2024–26 path)	EV per attempt (two runs)	EV per month (two runs)	Pass both
Random direction (version 2)	\$1,802 / \$1,928	\$3,587 / \$3,755	30%
Trend, 20 hours: direction of the last 20 hourly closes	\$885 / \$869	\$2,074 / \$2,024	30%
Trend, 5 hours: direction of the last 5 hourly closes	\$2,733 / \$2,801	\$4,613 / \$4,740	35–36%
Reversal, 5 hours: against the last 5 hourly closes	\$1,064 / \$1,110	\$2,327 / \$2,427	25%

Two runs of 6,000 attempts with different random start dates. On a synthetic zero-edge path the 20-hour trend rule gave $\$1,107 \pm 142$ per attempt against $\$1,282 \pm 156$ for random: the same within noise.

The spread is large: from half to one and a half times the random baseline. It is not evidence that any rule works. Every attempt that starts in the same week trades the same signals on the same two years of prices, so the 6,000 attempts behind each row are far from independent. On a synthetic zero-edge path, the trend rule and the random rule came out within noise of each other. One price history can make any rule look good or bad. The plan's figures rest on the synthetic paths, where by construction no rule can help.

WHAT TO DO WITH THIS

Pick one simple rule, write it down, apply it identically on every account, and plan on the zero-edge figures. If after several hundred trades your rule's full-size win rate at 1 : 5 is clearly above 1 in 6 after costs, that is the first evidence of an edge; until then there is none.

PART 2

Execution

The plan that maximised expected value per month per person in Part 1, written step by step. It stays inside every firm's published rules and FTMO's 1–1.5% risk guidance. It uses one position at a time on the Nasdaq and grows by adding accounts and firms, never by adding risk.

The plan on one page

READ THIS FIRST

- All figures assume zero skill and that every firm pays every payout. Payouts are discretionary; one refusal usually ends that firm.
- The trades have no edge. About seven attempts in ten lose their whole fee; one slot is ahead after a month about one time in five, after a year about 96% (Chapter 29).

23.1 Accounts

1. Start with FTMO 2-Step: four 100K accounts (or two 200K), the \$400,000 cap.
2. After the first payouts arrive, add firms one at a time (Section 21.5): FundingPips 2-Step ($4 \times 100K$), The5ers High Stakes ($1 \times 100K$), FundedNext Stellar 2-Step ($3 \times 100K$), Alpha Capital Pro 10% ($3 \times 100K$). Add FXIFY and GFT only after their caps are confirmed in writing (Chapter 26).
3. Everything in your own name, one registration per firm. Never trade for or with anyone else.

23.2 Every trade, on every account

1. **Instrument:** Nasdaq 100 only (US100 / NAS100 CFD; micro Nasdaq futures, MNQ, at a futures firm).
2. **One position at a time** per account; the next one as soon as the last closes.
3. **Window:** Monday to Friday, 01:00–20:00 UTC. No new trade after 19:00 UTC Friday; be flat by 20:30 UTC Friday.
4. **Direction:** the shared rule (Section 25.3). If any of your accounts already holds a Nasdaq position, a new position takes the same direction. No account ever opposes another.
5. **Risk** = the smallest of 1.5% of the account size, (room above the floor – 0.1%), and (today's loss room – 0.2%).
6. **Stop** = $0.75 \times$ the hourly move (about 60–65 Nasdaq points in 2024–26), attached on entry and never moved.
7. **Target** = $5 \times$ the stop distance; but if $5 \times$ risk would pass the phase or cycle target, the take-profit is the distance left plus this trade's cost.
8. **News:** no new trade from 15 minutes before to 5 minutes after a high-impact US release. On funded accounts, be flat 10 minutes before it.

23.3 Every account

- Never abandon a live account (Section 3.3). At a phase target, stop; fill missing minimum days with one tiny trade a day.
- **Funded:** each cycle, trade until +10% (or the account ends), then stop. Request the payout on the first eligible day and withdraw all profit (Chapter 28).
- Respect each firm's extra rules (Chapter 26).

Before you start

24.1 Requirements

- Adult (18+), and each firm's identity check in your own name.
- Money you can lose entirely, never borrowed. Because all accounts move together, the cash low point scales with the number of slots (Section 29.1): about \$15,000 typical and \$55,000 in a bad case (1 in 20) for four FTMO slots; about \$77,000 typical and \$165,000 bad for 23 slots.
- A platform per firm (MetaTrader 5, cTrader, DXtrade or the firm's own; NinjaTrader or Tradovate for futures), and either an automated program or several hours a day to place trades (Chapter 27).
- Each firm's current objectives, forbidden practices and payout terms, read in full and saved as PDF with the date.

24.2 Measure your costs at each firm

The plan rests on the Nasdaq costing about 2 points per round trip including slippage. At each firm, on a free demo:

1. Open the symbol specification. Write down the value of one point for one lot (V) and the minimum lot size.
2. During 14:00–20:00 UTC, note the spread every few minutes for a day and average it.
3. Add 0.5–0.6 points for slippage. If the total is above about 3 points, expect about a third less value at that firm; consider leaving it out.

24.3 Measure the hourly move

HOURLY MOVE (SIMPLE VERSION)

hourly move $\approx$ average of $|close - previous\ close|$ over the last 100 hourly bars $\times 1.25$

Recompute it every Monday and use the same value on every account for the week.

24.4 Copying trades between accounts

- **Within one firm:** use only a copying method the firm allows. FTMO allows copying between your own accounts within the cap. FundingPips allows copying between your own FundingPips accounts, but forbids inbound copying from signal providers or copier services. Check every other firm in writing.
- **Across firms:** place each firm's orders yourself, or run a separate program on each account that applies the same written rules. That is one trading plan applied independently on each account, not a copy service.
- Never copy trades from or to another person.

The trade, step by step

25.1 The formulas (any account size S)

RISK

$room = balance - floor$ $day\ room = daily\ limit - (start-of-day\ balance - current\ equity)$

$risk = \min(1.5\% \times S, room - 0.1\% \times S, day\ room - 0.2\% \times S)$

STOP, LOTS, COST

$stop\ points = 0.75 \times hourly\ move$ $lots = risk \div (stop\ points \times V)$ $cost\ \$ \approx (spread + 0.6) \times lots \times V$

TARGET

$to\ go = phase\ (or\ cycle)\ target\ balance - balance$

$target\ \$ = \min(5 \times risk, to\ go + cost\ \$ + \$5)$ $target\ points = target\ \$ \div (lots \times V)$

25.2 Worked examples (100K, hourly move 85 points, V = \$1, spread 1.4 points)

Situation	Balance	Room to floor	Day room	Risk	Lots	Cost	Target \$	Target pts	Win chance
Phase 1, start	\$100,000	\$10,000	\$5,000	\$1,500	23.43	\$47	\$7,500	320	17%
Phase 1, near target	\$106,500	\$16,500	\$5,000	\$1,500	23.43	\$47	\$3,552	152	30%
Phase 1, after a bad morning	\$95,400	\$5,400	\$1,400	\$1,200	18.75	\$38	\$6,000	320	17%
Phase 1, near the floor	\$90,900	\$900	\$5,000	\$800	12.50	\$25	\$4,000	320	17%
Phase 2, near target	\$104,000	\$14,000	\$5,000	\$1,500	23.43	\$47	\$1,052	45	59%

Win chance is the fair value, $stop \div (stop + target)$, before costs. Leverage does not bind: 23 lots at 31,000 is \$0.7 million notional, far below 1:50 on 100K.

25.3 The direction rule

SHARED DIRECTION

If any of your accounts holds a Nasdaq position: same direction as that position.

Otherwise: long if the last completed hourly close is above the close 5 hours earlier, short if it is below.

This is a standard short-term momentum rule, chosen because it is simple and common, not because it did best in Chapter 22's sample (choosing it for that reason would be fitting noise). It is not expected to have an edge. Its purpose is to be a written, repeatable plan, applied identically everywhere, that never opposes itself.

25.4 Futures accounts (Topstep 50K)

Micro Nasdaq futures pay \$2 per point. Risk about \$950 per trade; stop 0.75 hourly moves (about 50 points at 25,000) is about \$100 per contract, so about 9 contracts, well under the 50-micro limit. Use 1 : 3 here, not 1 : 5: the Combine's best-day rule (\$1,500, half the \$3,000 target) caps a single day's win, and in simulation 1 : 3 did best. Funded (Express) accounts: stop for the day once the day is a qualifying winning day (\$150+), and request the maximum allowed payout as soon as five winning days are in.

Firm-by-firm rule sheet

Summarised from public sources in October 2026 (Appendix D). Rules change often. Before buying, check each line against the firm's own current terms, and get the cap confirmed by support in writing where marked.

Programme	Targets	Daily / max loss	Min days	Funded payouts	Fee refund
FTMO 2-Step	10% / 5%	5% / 10% static	4 per phase	every 14 days, 80% (90% with scaling)	1st payout
FundingPips 2-Step	8% / 5%	5% / 10% static	3 per phase	bi-weekly 80%; monthly 100%; on-demand 90%	4th reward
FundedNext Stellar 2-Step	8% / 5%	5% / 10% static	5 per phase	day 21, then every 14 days, 80%	1st reward; +15% of challenge profit
The5ers High Stakes	10% / 5%	5% / 10%	3 profitable days ($\geq 0.5\%$)	bi-weekly, 80% rising with scaling	check
Alpha Capital Pro 10%	10% / 5%	5% (balance) / 10% static	3 per phase	14-day cycle from first trade, 80%	not assumed
FXIFY Two-Phase	10% / 5%	5% / 10% (one source: 4% / trailing)	check	14 days 80% or 30 days 100%	1st payout
GFT 2-Step Standard	10% / 5%	5% / 10% static	check	bi-weekly, 80%	refundable
Topstep 50K	\$3,000	\$2,000 end-of-day trailing	2 (best day $\leq$ \$1,500)	5 winning days of \$150+; $\leq 50\%$ of balance, $\leq$ \$2,000; 90%	— (\$49/month + \$149 activation)

Programme	Weekend (funded)	News (funded)	Consistency / caps	Copying
FTMO 2-Step	no (Standard)	± 2 min high impact	none	own accounts, within cap
FundingPips 2-Step	no (Master)	± 5 min restricted items	35% on on-demand and monthly	own FundingPips accounts; no inbound copier services
FundedNext Stellar 2-Step	check	allowed (check funded terms)	none	check
The5ers High Stakes	allowed	± 2 min high impact	none	check
Alpha Capital Pro 10%	no (funded Pro)	± 2 min high impact	40% for on-demand only	check; lot limit 40 on 100K
FXIFY Two-Phase	allowed	allowed	none (30% only on Lightning)	check
GFT 2-Step Standard	check	check	funded: $\leq$ \$3,000 profit/day; first 2 payouts $\leq 6\%$	check
Topstep 50K	futures hours	check	50% in the Combine	trade copier across own accounts

"check": not found in a reliable public source; ask the firm before buying. The plan already avoids weekends and news on every account, so those columns rarely bind.

THE RULES THAT APPLY EVERYWHERE

- No opposite positions across any of your accounts, at any firm, at any time.
- No accounts in other people's names, no third-party account management, no buying signals.
- Stay under each firm's allocation cap. FTMO suspends identically traded accounts whose total exceeds \$400,000.
- No exploiting platform errors, latency or price feeds; no tick scalping; no trades that exist only to meet a day count.
- Respect each firm's news windows and weekend rules on funded accounts.

Running many accounts: the daily routine

27.1 The work

At 1 : 5 a full-size trade lasts about four to eight hours including waiting, so each account takes one to three new trades a day. Accounts in the same direction usually open together. With 23 accounts that is 25–70 order tickets a day by hand, each sized for its own balance, or one program per account.

27.2 What a program does on each account

Step	Action
1	If this account has a position open, do nothing until it closes.
2	Outside 01:00–20:00 UTC Monday–Friday, after 19:00 UTC Friday, or within the news window: wait.
3	Read balance, start-of-day balance and the current phase or cycle target. If the target is reached, stop.
4	Compute risk (25.1). If the day room is below 0.3% of the account, wait for the next day.
5	Read the shared direction (25.3) from a common file or service that all your programs read.
6	Compute stop, lots, cost and target. Send a market order with both attached.
7	Log: time, account, balance, risk, lots, stop, target, direction, fill, result, cost.
8	Friday 20:30 UTC: close any open position. Funded: close 10 minutes before a listed release.

Check each firm's terms on automated trading before running a program there; FTMO permits Expert Advisors, Apex does not allow automation. Run every program on a free demo for two weeks first: both orders attached, sizes as in 25.2, the daily guard, the news rule and the Friday close all firing.

27.3 Weekly

- Monday: update the hourly move; check every firm's announcements for rule changes.
- Reconcile the log with each platform's statements; record every fee and payout.
- Run the checks of Section 29.3.

Funded accounts and payouts

28.1 Cycles of +10%

On every funded CFD account, trade the normal plan with the cycle target at +10% of the account size. When it is reached, stop; request a payout of all profit on the first allowed day. The balance returns to the start, the floor stays, and the next +10% cycle begins. GFT: aim for +7% in the first cycle (its first two payouts are capped at 6%).

About one cycle in two ends at +10% and one in two ends the account; that is expected. The total paid averages the split times the allowance (Chapter 5); +10% cycles collect it fastest.

28.2 What each firm's funded account delivers

Programme	Attempts that reach funded	Paid per funded account (average)	Funded account lasts (days, average)	First payout (day, median)
FXIFY Two-Phase	30%	\$7,200	16	22
FundingPips 2-Step, bi-weekly 80%	33%	\$7,200	17	24
The5ers High Stakes	30%	\$7,300	16	24
FundedNext Stellar 2-Step	34%	\$8,100	19	35
FundingPips 2-Step, monthly 100%	34%	\$8,800	30	40
Alpha Capital Pro 10%	30%	\$7,100	17	24
FTMO 2-Step	29%	\$7,300	16	26
FTMO 1-Step	33%	\$6,800	16	24
GFT 2-Step Standard	30%	\$6,800	25	22

28.3 Milestones of one FTMO attempt

Milestone, FTMO 2-Step, from the day you buy	Typical day
Phase 1 decided (pass about 46%)	day 4 (middle half 3–5)
Phase 2 decided (pass about 65% of those who reach it)	day 10.5
Funded account issued	day 11
First payout received, fee refunded (47% of funded accounts)	day 26 (middle half 25–27)
Funded account ends	day 20 (middle half 14–33)

With +10% cycles about half of funded accounts end before their first payout, fee unrefunded: the price of speed. If a dependable refund matters more, run the first cycle to +\$1,000 (91% reach it) at a cost of about 10% of the value per month (Section 19.3).

Month by month: cash, odds and when to stop

29.1 What to expect

Portfolio	Months	Chance ahead	Median cash	Bad case (1 in 20)	Good case (1 in 20)
FTMO only	1	19%	-\$10,100	-\$17,700	\$32,000
	3	75%	\$16,800	-\$37,900	\$120,400
	6	87%	\$68,200	-\$28,700	\$203,800
	12	96%	\$167,500	\$17,600	\$343,600
Seven CFD firms	1	36%	-\$15,700	-\$62,700	\$56,400
	3	91%	\$180,900	-\$26,700	\$434,500
	6	97%	\$492,000	\$50,600	\$999,900
	12	98%	\$1,101,900	\$243,700	\$2,090,000

Cumulative cash (payouts minus fees), all slots moving together. Typical cash low point: -\$15,200 (FTMO only) and -\$76,900 (23 slots); bad case -\$55,600 and -\$164,300.

The first month is almost always negative: fees are paid at once, payouts come four weeks later. The low point is usually in weeks three to six. Because the accounts move together, more slots multiply the swing as well as the expected value.

29.2 Bankroll rules

- Hold the "bad case" low point of Section 29.1 for the slots you run, in cash you can lose.
- Add firms only from profits already received, in the order of Section 21.5.
- Withdraw payouts to a separate account. Only the planned budget buys new attempts.
- Fix in advance how many months you will run before reviewing.

29.3 Checks that the assumptions hold

Check	Expected	Stop and review if
Win rate of full-size 1 : 5 trades (after 600)	≈ 16.7% before costs	below 13%
Average cost per trade (spread, slippage, financing)	≈ 3–4% of the risk	above 5.5%
Phase 1 pass rate (after 20 attempts at 10% targets)	≈ 47%	4 or fewer passes

Check	Expected	Stop and review if
Payouts honoured	all	any refusal: stop at that firm, read the stated reason, review the others
Rule changes	none	any change to caps, payouts, news or copying rules: re-check before the next purchase

29.4 Tax

Payouts are usually taxable income; fees are often deductible. With several firms, keep a single ledger of every fee and payout with dates and currencies, and check your country's rules with an adviser.

Final checklist

- ☐ I am 18 or older and hold, in cash I can lose, the bad-case low point for the slots I will run.
- ☐ I have read each firm's current objectives, forbidden practices, payout terms and cap, and confirmed the cap in writing where needed.
- ☐ I measured Nasdaq costs (≤ 3 points round trip) and the point value at every firm.
- ☐ My program or routine follows Chapter 23 exactly and passed two weeks on demo.
- ☐ One position per account; risk never above 1.5%; stop and target attached on entry; 1 : 5.
- ☐ All accounts use the same direction at the same time; none ever opposes another.
- ☐ The daily guard, news rule and Friday close are in place on every account.
- ☐ Funded accounts run +10% cycles and request payouts on the first eligible day.
- ☐ Everything is in my own name; no copying from or to anyone else.
- ☐ I understand that about seven attempts in ten lose the fee, that the first month is usually negative, and that firms may refuse payouts or change rules.

THE WHOLE DOCUMENT IN SIX LINES

A zero-skill trade is a fair bet, so a phase passes with probability $B \div (A + B)$. A funded account pays out, on average, the split times its loss allowance. Time is set by the instrument's cost per unit of movement, the stop, the reward-to-risk ratio and the firm's calendar. The best plan found is one Nasdaq position at a time, stop 0.75 hourly moves, 1 : 5, risk 1.5%, funded cycles of +10%. That gives about \$4,000 per month per 100K slot and, across the caps of seven CFD firms, about \$102,400 per month per person in theory. More positions per account do not help; more accounts do. The firms' discretion is the one thing the maths cannot price.

Formula sheet and glossary

Quantity	Formula	FTMO 100K example
Zero-skill win probability	$p = S \div (S + T)$	1 : 2 → 33.3%
Pass probability, static floor	$P = B \div (A + B)$	Phase 1: 50%; Phase 2: 66.7%
Pass probability from balance x	$P(x) = (x + B) \div (A + B)$	x = -\$5,000: 25%
With overshoot	$P = (B + O_{\text{floor}}) \div (A + O_{\text{target}} + B)$	±\$4,800 fixed: 41%
Intraday trailing, continuous prices	$P = e^{-A/D}$	A = 1.5D → 22.3%
Funded identity	Withdrawn = trading P&L + (start – final balance)	E[withdrawn] = \$10,000
Funded value	$\approx \text{split} \times B \times (1 - \text{cost share}) + \text{refund} \times q$	≈ \$8,600
First-payout probability	$q = B \div (B + h)$	h = \$500 → 95.2%
EV per attempt	$-\text{Fee} + \Pi P_i \times (\text{funded value} - \text{activation})$	+\$2,235 (no costs)
Cost ratio	$\kappa = \text{cost pips} \div \text{stop pips}$	0.75 ÷ 55 = 1.4%
Expected trades per phase	$A \times B \div L^2$	L = \$1,000: 100
Expected cost per phase	$\kappa \times A \times B \div L$	L = \$1,000: ≈ \$1,400
Expected trades per phase (1 : k)	$A \times B \div (k \times L^2)$	1.5%, 1 : 3: ≈ 15
Duration of one bracket	$\approx k \times s^2 \div \sigma^2$	s = 0.75σ, 1 : 3: ≈ 1.7 active hours
Phase time and cost	$\propto (s \div L)^2$ and $\propto c \div (k L s)$	Chapter 8
Instrument efficiency	weekly move ÷ round-trip cost	US100 481, EURUSD 147, BTC 49–63
Tightest stop from leverage λ	$L \div (\lambda \times \text{size})$	1.5% at 1:50: 0.03%
EV per month per slot	EV per attempt ÷ attempt length	Fast plan ≈ \$2,400
Kelly bankroll for one attempt	$W \approx \sigma^2 \div \mu$	≈ \$4,100–6,000
Lots (EURUSD)	risk ÷ (stop pips × \$10)	\$1,000, 60 pips → 1.66

Glossary

Bold play	Sizing each bet so a win lands exactly on the target, using as few bets as possible.
Bracket	A trade with stop-loss and take-profit attached on entry.
Consistency rule	A limit on how much of total profit one day may contribute.
EOD trailing drawdown	A floor that rises with the highest end-of-day balance.

Expected value (EV)	The average result per repetition over many repetitions.
Fair bet / martingale	A bet with zero expected value; a sequence of such bets.
Floor	The balance at which the account is closed (\$90,000 for FTMO 100K).
Gambler's ruin	The problem of reaching one level before another; answer $B \div (A + B)$.
Intraday trailing drawdown	A floor that follows the highest live equity, including open profit.
Kelly criterion	The stake that maximises long-run growth of wealth.
Lot	Position unit; 1 standard EURUSD lot = 100,000 EUR, \$10 per pip.
Optional stopping theorem	A fair game stopped by a reasonable rule is still fair on average.
Overshoot	Ending beyond a barrier instead of exactly on it.
Pip	0.0001 in EURUSD.
Profit split	Your share of withdrawn profit (80% at FTMO).
Slippage	Being filled at a worse price than your order.
Spread	The gap between the buy and sell price.
Static drawdown	A floor fixed relative to the starting balance.

Every account type modelled

Zero skill, best sizing for each rule set, payouts honoured. "EV" uses 1% trading costs; "1%-risk EV" restricts every trade to 1% of the account (25% of the drawdown for futures). CFD accounts are 100K, futures 50K, prices in USD (EUR at 1.17). Rules as published September–October 2026. † marks rule sets partly inferred; instant-account values ignore payout caps.

Firm	Account	Price	Target / max loss	Funded	P(fund.)	Value	EV	1% risk
FTMO	1-Step	\$584	10/10% EOD trail +cons.	90% × \$10k	43%	\$9,376	+\$3,465	+\$2,646
Blue Guardian	Instant	\$779	instant	80% × \$6k	100%	\$3,920	+\$3,141	+\$3,062
Goat Funded	Instant GOAT	\$838	instant	80% × \$6k	100%	\$3,920	+\$3,082	+\$3,002
The5ers	High Stakes 2-Step	\$545	8/10% static → 5/10% static	80% × \$10k	36%	\$8,359	+\$2,490	+\$2,054
FundedNext	Stellar 2-Step	\$550	8/10% static → 5/10% static	80% × \$10k	36%	\$8,364	+\$2,486	+\$2,056
FundingPips	2-Step	\$529	8/10% static → 5/10% static	80% × \$10k	36%	\$8,281	+\$2,477	+\$2,043
Alpha Capital †	Alpha Pro 10%	\$497	10/10% static → 5/10% static	80% × \$10k	33%	\$8,313	+\$2,215	+\$1,794
FTMO	2-Step	\$632	10/10% static → 5/10% static	80% × \$10k	33%	\$8,442	+\$2,122	+\$1,697
Goat Funded †	2-Step Standard	\$594	10/10% static → 5/10% static	80% × \$10k	33%	\$7,840	+\$1,964	+\$1,556
FundedNext	Stellar Lite	\$400	8/8% static → 4/8% static	80% × \$8k	33%	\$6,272	+\$1,646	+\$1,398
Blue Guardian †	2-Step Pro	\$464	10/10% EOD trail → 10/10% EOD trail	85% × \$10k	24%	\$8,330	+\$1,556	+\$597
Maven †	2-Step	\$396	8/8% static → 5/8% static	80% × \$8k	30%	\$6,272	+\$1,490	+\$1,252
Alpha Capital †	Alpha Pro 8%	\$577	8/8% static → 5/8% static	80% × \$8k	30%	\$6,815	+\$1,473	+\$1,222
FundedNext	Stellar 1-Step	\$570	10/6% static	90% × \$6k	37%	\$5,292	+\$1,374	+\$1,194
FundingPips	1-Step	\$555	10/6% static	80% × \$6k	37%	\$5,120	+\$1,326	+\$1,158
FundingPips	Zero (instant)	\$499	instant	95% × \$5k	100%	\$1,679	+\$1,180	+\$1,113
E8 Markets †	E8 One (6% DD)	\$464	9/6% EOD trail	80% × \$6k	39%	\$3,920	+\$1,079	+\$514
Goat Funded †	3-Step GOAT	\$365	6/8% static → 6/8% static → 6/8% static	80% × \$8k	18%	\$6,272	+\$767	+\$598
FundingPips	2-Step Pro	\$399	6/6% static +cons. → 6/6% static +cons.	80% × \$6k	24%	\$4,704	+\$726	+\$635
Maven †	3-Step	\$269	3/3% static → 3/3% static → 3/3% static	80% × \$3k	12%	\$2,352	+\$14	+\$3
Alpha Capital †	Alpha Three	\$397	8/6% static → 8/6% static → 8/6% static	80% × \$6k	8%	\$4,704	−\$39	−\$95

Firm	Account	Price	Target / max loss	Funded	P(fund.)	Value	EV	1% risk
Lucid	Direct 50K (instant)	\$520	instant	90% × \$2k	100%	\$1,458	+\$938	+\$908
Apex	50K EOD	\$58 +\$139	\$3k/\$2k EOD trail	100% × \$2k	39%	\$1,620	+\$525	+\$352
Bulenox	50K Opt.2 (EOD)	\$175/mo +\$143k	\$3k/\$2.5k EOD trail +cons.	100% × \$2.5k	36%	\$2,028	+\$502	+\$482
Lucid	Flex 50K	\$130	\$3k/\$2k EOD trail	90% × \$2k	40%	\$1,458	+\$446	+\$274
Tradeify	Growth 50K	\$139	\$3k/\$2k EOD trail	90% × \$2k	39%	\$1,458	+\$435	+\$265
Tradeify	Lightning 25K (instant)	\$329	instant	90% × \$1k	100%	\$723	+\$394	+\$379
Lucid	Pro 50K	\$185	\$3k/\$2k EOD trail	90% × \$2k	39%	\$1,458	+\$389	+\$219
Topstep	Combine 50K (Standard)	\$49/mo +\$143k	\$3k/\$2k EOD trail +cons.	90% × \$2k	32%	\$1,470	+\$375	+\$316
Topstep	Combine 50K (No-Activation)	\$109/mo	\$3k/\$2k EOD trail +cons.	90% × \$2k	32%	\$1,470	+\$363	+\$298
My Funded Futures	Flex 50K	\$107/mo	\$3k/\$2k EOD trail +cons.	80% × \$2k	32%	\$1,296	+\$309	+\$252
Tradeify	Select 50K	\$159	\$3k/\$2k EOD trail +cons.	90% × \$2k	29%	\$1,458	+\$270	+\$245
The5ers †	Futures Basecamp 50K	\$100	\$3k/\$1.5k static	80% × \$1.5k	33%	\$980	+\$222	+\$202
Apex	50K Intraday	\$32 +\$99	\$3k/\$2k intraday	100% × \$2k	22%	\$672	+\$92	+\$86
Bulenox	50K Opt.1 (intraday)	\$175/mo +\$143k	\$3k/\$2.5k intraday	100% × \$2.5k	29%	\$848	+\$31	+\$21
Take Profit Trader	PRO 50K	\$170 +\$130	\$3k/\$2k EOD trail	80% × \$2k	40%	\$565	+\$2	-\$53
My Funded Futures	Rapid 50K	\$157/mo	\$3k/\$2k intraday	90% × \$2k	22%	\$604	-\$26	-\$31

Method and sources

How the numbers were produced

- **Exact results** (Chapters 3–6) follow from the formulas shown and were reproduced by a coin-flip simulation of 200,000 attempts (EV +\$2,272 ± 31 against the exact +\$2,235).
- **Evaluation phases for other firms** were solved by dynamic programming over balance and end-of-day high-water mark, with daily loss limits, best-day caps and a cost of $\kappa\sqrt{wl}$ per bracket.
- **Funded values** use the identity of Chapter 5 with lock levels and payout caps.
- **Real-data tests** use hourly EURUSD (yfinance), 18 Dec 2023 – 2 Oct 2026, as described in Chapter 14.

Sources

- FTMO, Trading Objectives: ftmo.com/en/trading-objectives
- FTMO, Forbidden Trading Practices: ftmo.com/en/forbidden-trading-practices
- FTMO, 1-Step Challenge: ftmo.com/en/1-step-challenge
- Topstep, Express Funded Account parameters: help.topstep.com
- PropFirmsFinder program pages (FTMO, Topstep, Apex, The5ers, FundedNext, FundingPips, E8 Markets, Alpha Capital, Blue Guardian, Maven, Goat Funded Trader, Tradeify, Lucid, Take Profit Trader, My Funded Futures, Bulenox): propfirmsfinder.com
- Apex Trader Funding payout caps: thepropjournalist.com/reviews/apex-trader-funding; PA rules: traderssecondbrain.com
- Tradeify, Lucid, Take Profit Trader, FundedNext rule guides: tradetanto.com, proptradingvibes.com
- Pass-rate estimates: traderssecondbrain.com/guides/prop-firm-pass-rate
- Prop firm closures: tradernotion.com; fxnewsgroup.com (True Forex Funds)

Rules and prices change frequently. Every figure is a snapshot from October 2026 and should be re-checked against the firm's current terms.

Version 3: firm rules and sources

Firm rules, caps and prices were collected in October 2026 from the firms' own pages where possible and otherwise from review sites. Prices change with promotions; the fee used is shown in Chapter 20. Check every item against the firm's current terms before buying.

Topic	Source
FTMO: account limit, \$400,000 cap, identical strategies	ftmo.com/en/faq/how-many-accounts-can-i-have/
FTMO: monitored behaviour (gambling, all-in, no fixed risk limit, 1–1.5% guidance)	ftmo.com/en/blog/why-ftmo-monitors-certain-patterns-in-trading-behaviour/
FTMO: 1-Step and 2-Step rules, news and weekend rules	propfirmbriefing.com/prop-firm-rules/ftmo/
FundedNext: \$300,000 funded cap, evaluations unlimited	help.fundednext.com (How many accounts can I have with FundedNext?)
FundedNext Stellar 2-Step: targets, price, 21-day first reward, refund, 15% challenge share	fundednext.com/cfds/stellar-2-step ; help.fundednext.com ; proptradingvibes.com/blog/fundednext-stellar-2-step
FundingPips: models, payouts, refund with 4th reward, \$400,000 cap incl. evaluations, copying and weekend rules	proptradingvibes.com/blog/fundingpips-rules ; fundingpips.com/blog/fundingpips-maximum-allocation
The5ers High Stakes: targets, profitable days, prices, account limits	the5ers.com/high-stakes/ ; propfirmbridge.com (High Stakes review); tradetanto.com
Alpha Capital: programmes, lot limits, news and weekend rules, \$400,000 / \$300,000 caps, price	tradetanto.com/learn/alpha-capital-group-rules-what-traders-need-to-know ; propvator.com/blog/alpha-capital-max-capital-allocation/
FXIFY: programmes, consistency rule (Lightning only), cap	tradingfinder.com/props/fxify/rules/ ; propvator.com/blog/fxify-max-capital-allocation/
Goat Funded Trader: 2-Step Standard, funded caps, \$400,000 cap, price	propvator.com/blog/goat-funded-trader-max-capital-allocation/ ; review summaries (bestprop.io , cryptoslate.com)
E8 Markets: \$4.85M household cap (not modelled: rules too product-specific to verify)	propvator.com/blog/e8-markets-max-capital-allocation/
Apex (after the 2026 update): prices, drawdowns, payouts, 20 accounts, no automation	proptradingvibes.com/blog/apex-trader-funding-rules-overview ; proptradingvibes.com/blog/apex-trader-funding-payout-rules
Topstep: Combine, Express Funded payouts, \$49 + \$149, 5 accounts	proptradingvibes.com/blog/topstep-trading-combine-rules ; proptradingvibes.com/blog/topstep-pricing-breakdown ; topstep.com
Tradeify Select and Lucid Flex: rules, prices, payouts	phidiaspropfirm.com/education/tradeify-select ; proptradingvibes.com/blog/lucid-trading-lucidflex-account

D.1 Method in one paragraph

Synthetic paths: 12 years of hourly bars, zero-drift random walk at the Nasdaq's hourly volatility (0.27%), exact Brownian-bridge extremes, weekends closed. Costs: 0.0064% of notional per round trip (US100 CFD), 0.0027% (micro futures), plus financing. 24,000 synthetic and 8,000 real-path attempts per firm; portfolios from 2,000 twelve-month histories per firm.